

Reconciling the Investment-Uncertainty Relationship

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Abstract

Does uncertainty increase or decrease investment? Empirical evidence remains decidedly mixed, both in the time-series and in the cross-section. I present a dynamic model of investment that unifies these seemingly conflicting facts. Central to the analysis, I use a new statistic—the distance to investment—which measures how close a project is to having a positive net present value. I derive closed-form expressions for optimal investment under stochastic volatility and find three main results. (1) Over a fixed time horizon, high average uncertainty is associated with higher investment if the initial distance is “far”, but lower investment if the distance is “close”. (2) Uncertainty shocks increase investment only when they coincide with higher expected growth rates. (3) In general equilibrium, an uncertainty shock can raise investment during economic contractions, contrary to the partial-equilibrium intuition.

Keywords: Uncertainty, Investment, Horizon, Economic Growth

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1 Introduction

The relationship between investment and firm-level uncertainty is central to economics and finance. Uncertainty—defined by the volatility of payoffs—is a key ingredient in the firm’s price of risk, which in turn drives capital budgeting and planning decisions. But despite the importance of this topic, the relationship between the two in the data is unclear.

In the time-series, a large body of work shows that uncertainty shocks depress subsequent capital investment, notably [Bloom 2009](#), [Gulen and Ion 2016](#), [Alfaro et al. 2024](#) among many others. This is consistent with the “bad-news principle” of [Bernanke 1983](#) and the classic real-options view that high volatility raises the option value of waiting. At the same time, [Atanassov et al. 2024](#) document a *positive* impact of political uncertainty on subsequent research and development. The authors argue that their findings appeal to a time-to-build view of investment in the spirit of [Bar-Ilan and Strange 1996](#).

Evidence in the cross-section is equally divided. [Leahy and Whited 1996](#) and [Kermani and Ma 2023](#) document a negative relationship between uncertainty and investment in their regressions of investment on firm-level risk, arguing in favor of models of irreversible investment ([Dixit and Pindyck 1994](#)). On the other hand, [Chang et al. 2024](#) documents a positive relationship in support of their model and [Abel 1983](#). [Kang 2025](#) even finds that the relationship can change depending on the horizon at which uncertainty is measured.

These conflicting facts raise fundamental questions about the underlying economic mechanism. One natural interpretation is that the sign depends on measurement choices: quarterly versus annual investment rates, equity versus asset volatility, physical versus intangible capital or differences in econometric specification. But it is unclear how these choices matter theoretically, nor is there a unifying framework that can rationalize both a positive and negative relationship; existing models take a stance on a single sign.

This paper proposes a unified framework that reconciles these conflicting empirical findings. I show that a single economic mechanism, the distance to investment, organizes both cross-sectional and time-series variation in investment responses to uncertainty. The distance to investment is simply a statistic that measures how close a project is to having a positive net present value; it determines whether a firm is “far” or “close” from exercising an option to invest. The role of volatility is straightforward: it affects the likelihood at which an unprofitable project becomes profitable and vice-versa.

Intuitively, suppose, for example, the current price of output is \$1.50 but investment becomes profitable only once the price reaches \$2. When the price lies below the threshold, the distance to investment is large. In this region, increased volatility is beneficial: large price movements accelerate the probability of crossing the threshold, raising investment over any fixed horizon. In contrast, if the current price is \$2.50—above the threshold—and the drift of prices is positive, then high volatility increases the chance of falling below it. Here,

volatility hurts investment when projects are nearly or already profitable. This single statistics delineates the conditions under which uncertainty encourages or discourages investment. In the cross-section, firms that face a larger share of projects that are initially unprofitable gain more from volatility, whereas firms with projects close to profitability do not.

To formalize the mechanism, I develop a tractable dynamic model of irreversible investment (Section 2.1) in which new investment projects arrive over time. Each project consists of a bundle of factor inputs, incurs a fixed cost, and expires if not immediately adopted. With this set-up, the firm solves a sequence of optimal stopping problems. I derive closed-form solutions for the expected occupation time above the investment thresholds, a key statistic that govern investment intensity across horizons.

To reconcile the conflicting signs in the time-series, I then incorporate stochastic volatility in Section 2.3 using a two-state Markov regime-switching process, in which short bursts of high uncertainty is followed by prolonged periods of low uncertainty. I show that when the firm finds itself in the high uncertainty regime, its optimal investment decision includes a precautionary motive that increases distance. Because investment is irreversible and most time is spent in the low-volatility regime, the firm delays investment during high-uncertainty periods to avoid an unfavorable change to prices before conditions normalize. Consequently, uncertainty shocks typically depress investment. The only exception occurs when projects require time to become profitable and uncertainty shocks are accompanied by increases in expected growth rates.

In Section 2.4, I endogenize the firm’s cost of capital within a general-equilibrium framework, an important departure from the standard approach that assumes a constant discount rate. This is important since temporary shocks to aggregate uncertainty should also be accompanied by changes to the market price of risk; in contrast, most of the literature uses a constant discount rate. Importantly, the investment threshold for a given firm is a function of the equilibrium risk-free rate, which in turn is determined by expectations about future aggregate investment. One key implication is that the impact of uncertainty shocks varies systematically with the state of the economy. During economic slowdowns, an uncertainty shock can stimulate investment because the riskless rate is relatively low, thereby keeping investment thresholds low. As a result, high volatility may help firms reach their investment triggers sooner. In contrast, when the economy is overheating, an uncertainty shock compounded by a higher riskless rate raises investment thresholds too far, reducing investment.

Finally in Section 3, I measure the distance to investment empirically and find preliminary evidence for the model prediction in the cross-section. In particular, I regress physical capital investment on firm-level uncertainty, a proxy for distance, and an interaction term of the two. Consistent with the model, I find that the coefficient on the interaction term is positive while the coefficients on uncertainty and distance are negative. This supports the model

prediction in the cross-section that firms who naturally face a larger proportion of investment opportunities that are not immediately profitable benefit from higher uncertainty.

Together, these results provide a unified explanation for the conflicting empirical findings on uncertainty and investment. The distance to investment emerges as a useful statistic that organizes both time-series and cross-sectional variation, delivering clear conditions under which uncertainty stimulates or depresses investment.

1.1 Related Literature

The primary contribution of this paper is to provide a tractable framework that can explain the different signs in both the time-series and in the cross-section.

In particular, the literature on investment under uncertainty with constant volatility can be divided into models that predict a positive cross-sectional relationship between firm-level uncertainty and investment and ones that predict a negative relationship. In the former camp, the convexity of payoffs with respect to the stochastic variable leads to a positive relationship, starting from [Oi 1961](#), [Hartman 1972](#) and [Abel 1983](#), and as recently as [Chang et al. 2024](#). Other models that predict a positive relationship without relying on convexity include [Roberts and Weitzman 1981](#) and [Bar-Ilan and Strange 1996](#) who study sequential and multi-stage projects.¹ In the latter camp, non-convex investment costs lead to an investment “no-action” region. In the comparative statics, because an increase in σ expands the no-action region, authors then conclude that uncertainty delays investment ([Bernanke 1983](#); [Pindyck 1988](#); [Dixit and Pindyck 1994](#)).

My paper contributes to this literature by rationalizing both sides using a single statistic: the distance to investment, without changing the model assumptions along the way. To my knowledge, [Caballero 1991](#) also reconciles the conflicting signs in the cross-section, though the author requires perfect competition and symmetric adjustment costs for a positive sign, and imperfect competition and asymmetric adjustment costs for a negative one.

On the literature on investment under uncertainty with stochastic volatility, this paper is potentially the first to present a dynamic model that can rationalize an *increase* in investment in response to an uncertainty shock.

The closest paper to mine is likely [Bloom 2009](#). One of the key differences between our papers is that in [Bloom 2009](#), the author assumes that firms are close to the optimal exercise boundary before an uncertainty shock. Mechanically, an uncertainty shock will expand the

¹In [Roberts and Weitzman 1981](#), the firm has an option to abandon a project before completion. In their model, the firm learns about the terminal payoff progressively after completing a stage of the project. When uncertainty is high, completing early stages is valuable due to the high value of information gained. In this manner, high uncertainty can “increase” investment. But because the focus of their work is on the optimal exploration and development for different type of processes, it is unclear if their results translate to average effects over multiple projects.

no-action region, which leads to an immediate delay in investment before catching up due to the drift of stochastic process. In contrast, the distance to investment statistic is defined by how far the optimal exercise boundary is from the current level of the state variable. Hence, explaining the finding that an uncertainty shock can increase investment comes from this difference.

Moreover, my paper endogenizes the firm's cost of capital in general equilibrium whereas the cost of capital in [Bloom 2009](#) is a constant, regardless of the volatility regime. In fact, this omission typically holds for most papers in this literature. Consequently, I show that a decrease in investment in response to an uncertainty shock occurs when the economy is growing; on the other hand, when the economy is contracting, my model predicts that an uncertainty shock can boost investment.

2 Model

I first outline the setup of the model in Section [2.1](#). To build intuition, in Section [2.2](#), I derive the firm's optimal investment policy subject to an exogenous stochastic discount factor in the economy in which the firm faces a constant level of uncertainty. In Section [2.3](#), I add stochastic volatility to this framework. Finally, in Section [2.4](#), I derive the general equilibrium results.

2.1 The Setup

In this section, I derive the firm's optimal investment policy subject to an exogenous stochastic discount factor in the economy. I first focus on the case in which the firm faces a constant level of uncertainty, then add stochastic volatility into the framework.

2.1.1 Technology

Time is continuous. Consider a competitive firm that produces consumption goods in each period with a constant returns to scale and constant elasticity of substitution (CES) production technology,

$$y(t, A) = \left[\alpha(z_t)^\varepsilon + (1 - \alpha)(k_t)^\varepsilon \right]^{\frac{1}{\varepsilon}} A_t^\theta \quad (1)$$

where A_t is a stochastic process that drives the productivity of output,

$$dA_t = A_t \mu dt + A_t \sigma dB_t$$

and B_t is a standard Brownian motion. In equation [\(1\)](#), the parameter α controls the relative share while ε relates to the elasticity of substitution between the two production inputs (complements at $\varepsilon = 0$, substitutes at $\varepsilon = 1$). The parameter θ controls the curvature

of output with respect to the stochastic variable.² Throughout this paper, I consider the linear case, $\theta = 1$.

Importantly, z_t and k_t denote two types of capital inputs allocated to the production of the good. For simplicity, I assume production does not require labor. Moreover, to facilitate the derivation of closed-form expressions, I make the following assumption regarding capital accumulation in this model,

Assumption 1. An investment project is a bundle of inputs $(\tilde{z}_t, \tilde{k}_t)$. The firm can acquire this bundle at a fixed cost q . Once adopted, production expands from $[\alpha(z_t)^\varepsilon + (1 - \alpha)(k_t)^\varepsilon]^\frac{1}{\varepsilon}$ to $[\alpha(z_t + \tilde{z}_t)^\varepsilon + (1 - \alpha)(k_t + \tilde{k}_t)^\varepsilon]^\frac{1}{\varepsilon}$.

Assumption 2. With probability 1, an investment project arrives in each unit of time. The *composition* of the investment projects are independent random draws from uniform distributions on $[\underline{\tilde{z}}, \bar{\tilde{z}}]$ and $[\underline{\tilde{k}}, \bar{\tilde{k}}]$ at the time of arrival. Moreover, a project expires if it is not adopted by the arrival of the next project.

Given the cost q , Assumption 1 simplifies the investment choice to an optimal stopping problem. Importantly, with two capital inputs and the CES aggregator, the marginal benefit of an investment project to the firm differs on a project-by-project basis. When the composition of the investment project closely mirrors the firm's existing capital stock, the project is not very valuable. As a result, due to Assumption 2, the firm may skip projects that do not meet its current needs. This conflict highlights the fact that a competitive firm may not always have access to its preferred type of capital; some projects are more beneficial than others.

2.2 Optimal Investment

In order to describe the firm's optimal investment policy, I make the following assumptions. Suppose markets are complete, Modigliani-Miller theorem holds, and firms discount cash flows at a constant rate r .

Next, I apply a change of variable from A_t to $x_t = \log A_t$. Moreover, for ease of notation, let $\zeta_n \doteq [\alpha(z_0 + \sum_n \tilde{z}_n)^\varepsilon + (1 - \alpha)(k_0 + \sum_n \tilde{k}_n)^\varepsilon]^{1/\varepsilon}$. The subscript n indicates the number of projects that are completed, where z_0 and k_0 are the initial capital stock and $\tilde{z}_n$ and $\tilde{k}_n$ are the n th realizations of the project draws. Finally, let $\tilde{\zeta}_{n+1}$ denote the case in which the composition of the $n + 1$ th project (which will arrive in the future) uses expected values of

²In many models with Cobb-Douglas production (such as [Abel 1983](#)), the adjustability of labor input leads indirect utility to be a convex function of the stochastic variable. Here, I assume that labor is not required in the production of the good. For the purposes of comparison, the parameter θ allows me to incorporate convexity into the model.

the random draws. The Hamilton-Jacobi-Equation (HJB) equation can be written:

$$rV(\zeta_n, x) = \zeta_n e^x + (\mu - 0.5\sigma^2)V_x(\zeta_n, x) + \frac{1}{2}\sigma^2 V_{xx}(\zeta_n, x) + [V(\tilde{\zeta}_{n+1}, x) - V(\zeta_n, x)]$$

with boundary conditions:

$$V(\zeta_n, \hat{x}) = V(\zeta_{n+1}, \hat{x}) - q$$

where $\hat{x}$ denotes the threshold value at which the firm adopts today's project. Since projects instantly expire, the decision rule is to accept projects with a positive net present value (NPV). The optimal threshold $\hat{x}$ at which projects are positive NPV is given by

$$\hat{x} = \log \left(\frac{\varphi^+}{\varphi^+ - 1} \frac{q}{\frac{1}{1+\delta} \left[\zeta_{n+1} - \zeta_n + \frac{\tilde{\zeta}_{n+2}}{\delta} - \frac{\tilde{\zeta}_{n+1}}{\delta} \right]} \right)$$

where $\varphi \pm = -(\mu - .5(\sigma^i)^2)/(\sigma^i)^2 \pm \sqrt{2(1+r)(\sigma^i)^2 + (\mu - .5(\sigma^i)^2)^2}/(\sigma^i)^2$. The above equation has an intuitive interpretation. The threshold level at which the firm invests is a ratio of the investment cost over the incremental gain in output from this project. The greater the marginal benefit, the lower the required threshold and the more likely the firm adopts it. In the denominator, the difference $\zeta_1 - \zeta_0$ reflects the immediate gain to production while the difference in the next two terms, $\frac{\tilde{\zeta}_{n+2}}{\delta} - \frac{\tilde{\zeta}_{n+1}}{\delta}$, reflects the incremental gain from future projects if today's project is adopted.

To transition from investment thresholds to the firm's investment *rate*, we can use the fact that if one project arrives at each point in time, the expected time the stochastic variable x_t spends outside of the investment region gives a natural characterization of the expected number of projects the firm adopts over a fixed time interval $[t, T]$. Define the expected occupation time,

$$f(t, x) = \mathbb{E}_t \left[\int_t^T \mathbb{1}_{x_s \geq \hat{x}_s} \mathbf{d}s \right] \quad (2)$$

which allows for the fact that $\hat{x}_t$ can change as the firm adopts more projects. Then, the investment rate is given by $I_T = f(t, x)/T$. The following lemma characterizes the investment rate to this setting.

Lemma 1. Fix T . Suppose $z_t/k_t = \tilde{\zeta}/\tilde{k}$. Furthermore, let $\bar{\mu} \doteq \mu - 0.5\sigma^2$. Then,

$$I_T = \frac{1}{T-t} \int_t^T \left[1 - \Phi \left(\frac{\hat{x} - x_0 - \bar{\mu}s}{\sigma\sqrt{s}} \right) \right] \mathbf{d}s \quad (3)$$

where x_0 is the initial position of x .

Lemma 1 uses the well-known formula for the density that $\Pr(x_s > \hat{x})$ by time s . More-

over, if the composition of the firm's existing assets mirrors the average composition of investment projects, due to the homogeneity of the production function, $\hat{x}$ is a constant in expectation even as the firm adopts more and more projects.

Though the formula in Equation 3 can be evaluated numerically without issue, we can derive more explicit representations by rewriting the expectations operator in Equation 2 in terms of a differential equation.

First, appealing to the Feynman-Kac formula, we can represent the conditional expectation $f(t, x)$ as a solution to the forward equation,

Lemma 2. $f(t, x)$ obeys the following partial differential equation:

$$\frac{\partial f(t, x)}{\partial t} + \left[\mu - \frac{1}{2}\sigma^2 \right] \frac{\partial f(t, x)}{\partial x} + \frac{1}{2}\sigma^2 \frac{\partial^2 f(t, x)}{\partial x^2} = -\mathbb{1}_{x > \hat{x}}$$

with the terminal condition $f(T, x) = 0$.

Proof. See Appendix 5.1. □

As always, it is convenient to convert the partial differential equation into an ordinary differential equation. Consider the change of variable $\tau = T - t$ and the Laplace transform, $\bar{f}(\xi, x) = \mathcal{L}\{f\} \doteq \int_0^\infty e^{-\xi\tau} f(\tau, x) d\tau$.³

The PDE then becomes the ODE:

$$-\xi \bar{f}(\xi, x) + \left[\mu - \frac{1}{2}\sigma^2 \right] \frac{\partial \bar{f}(\xi, x)}{\partial x} + \frac{1}{2}\sigma^2 \frac{\partial^2 \bar{f}(\xi, x)}{\partial x^2} = -\frac{1}{\xi} \mathbb{1}_{x > \hat{x}}$$

Proposition 1. Let $\bar{\mu} \doteq \mu - 0.5\sigma^2$. Given the current value of x , the expected occupation

³The Laplace transform has the following properties:

$$\mathcal{L}\left\{\frac{\partial f}{\partial t}\right\} = \int_0^\infty e^{-\xi\tau} \frac{\partial f}{\partial t} d\tau = -\xi \bar{f}(\xi, x) + f(0, x)$$

via the chain rule and integration by parts and

$$\mathcal{L}\left\{\frac{\partial f}{\partial x}\right\} = \int_0^\infty e^{-\xi\tau} \frac{\partial f}{\partial x} d\tau = \frac{\partial}{\partial x} \int_0^\infty e^{-\xi\tau} f d\tau = \frac{\partial \bar{f}}{\partial x}$$

via Leibniz's Rule. While Fourier Transforms have been used before in Finance, because the transform variable in this setting is time and not probability, the Laplace transform—which only integrates over the positive real line—is the appropriate transformation.

time over the time interval $\tau \doteq T - t$ is

If $x < \hat{x}$:

$$f(\tau, x) = \tau - \frac{\hat{x} - x}{\bar{\mu}} - \frac{1}{2} \frac{\sigma^2}{\bar{\mu}^2} + \frac{1}{2\pi} \int_0^\infty \frac{1}{(\frac{\bar{\mu}^2}{2\sigma^2} + \xi)^2} e^{-(\frac{\bar{\mu}^2}{2\sigma^2} + \xi)\tau - \frac{\bar{\mu}}{\sigma^2}(x - \hat{x})} \times \left[\sin\left(\frac{\sqrt{2\xi}}{\sigma}(\hat{x} - x)\right) + \frac{\bar{\mu}}{\sigma\sqrt{2\xi}} \cos\left(\frac{\sqrt{2\xi}}{\sigma}(\hat{x} - x)\right) \right] d\xi \quad (4)$$

If $x > \hat{x}$:

$$f(\tau, x) = \tau - \frac{1}{2} \frac{\sigma^2}{\bar{\mu}^2} e^{-\frac{2\bar{\mu}}{\sigma^2}(x - \hat{x})} + \frac{1}{2\pi} \int_0^\infty \frac{1}{(\frac{\bar{\mu}^2}{2\sigma^2} + \xi)^2} e^{-(\frac{\bar{\mu}^2}{2\sigma^2} + \xi)\tau - \frac{\bar{\mu}}{\sigma^2}(x - \hat{x})} \times \left[\frac{\bar{\mu}}{\sigma\sqrt{2\xi}} \cos\left(\frac{\sqrt{2\xi}}{\sigma}(x - \hat{x})\right) - \sin\left(\frac{\sqrt{2\xi}}{\sigma}(x - \hat{x})\right) \right] d\xi \quad (5)$$

Proof. See Appendix 5.2. □

Focusing on Equation (4) and the case $x < \hat{x}$ in which projects are currently ‘out-of-money’, the expected time a firm spends above its investment threshold over a horizon of τ years can be decomposed into a long-run ergodic component and a transient, oscillatory one. The first component, equal to $\tau - \frac{\hat{x} - x}{\bar{\mu}} - \frac{1}{2} \frac{\sigma^2}{\bar{\mu}^2}$, asks: given the current distance, how long will it take for the drift to carry the firm above its investment threshold? Since τ is the maximum time that the firm can spend in the investment region over τ years, the occupation time takes τ and subtracts the first passage time from x to $\hat{x}$ given $\bar{\mu}$, and a variance correction to account for the randomness of the stochastic process.

To interpret the second component—the remaining integral, it is important to note that the integral vanishes to 0 as $\tau \rightarrow \infty$.⁴ Moreover, since the function decays exponentially by $\bar{\mu}^2/(2\sigma^2)$, this implies that the function takes its largest values in absolute terms over short τ , then “resets” through oscillations of the sinusoidal terms. Taken together, we can interpret the second component as capturing the transitory effects of a Brownian shock on the expected occupation time via short-term excursions around the current state variable x before the long-run dynamics eventually take hold, decaying at a rate $\bar{\mu}^2/(2\sigma^2)$.⁵

Thus, one advantage of this decomposition over the probabilistic representation in Equation (3) is that it tells us the rate at which a firm’s current investment policy deviates from its long-run average is proportional to its squared Sharpe ratio. In fact, the squared Sharpe ratio

⁴More formally, we can see that term-by-term, the dominating convergence theorem applies and we can bring the limit as $\tau \rightarrow \infty$ inside the integral. Then, every term is multiplied by 0 which concludes the proof.

⁵As the proof in Appendix (5.2) shows, the decomposition of investment intensity into two components appears naturally, as the solution can be separated into the value of the residue at the pole at $\xi = 0$ and the branch cut contribution at $\xi = -\bar{\mu}^2/(2\sigma^2)$. Since the pole at $\xi = 0$ in the Laplace domain corresponds to a term that persists indefinitely in the time domain—it does not decay—, it is appropriate to consider the first component as the principal, ergodic behavior.

$\bar{\mu}^2/\sigma^2$ is a sufficient statistic for investment rate persistence; firms with high signal-to-noise in their productivity dynamics converge rapidly to ergodic behavior, exhibiting low persistence in investment rates in response to temporary shocks, whereas firms with relatively high variance carry the information content of their current position for many periods, generating high persistence in investment dynamics. This leads to the first testable hypothesis:

H1. The persistence of a firm's investment rate is decreasing in its squared Sharpe ratio.

The following corollary leads to the next testable prediction,

Corollary 1. Asymptotically as $\tau \rightarrow 0$,

If $x < \hat{x}$:

$$f(\tau, x) \sim \frac{2\sigma^3}{(\hat{x} - x)^3 \sqrt{2\pi}} \times \tau^{5/2} \times e^{-(\hat{x}-x)^2/(2\sigma^2\tau) + \bar{\mu}(\hat{x}-x)/\sigma^2} \quad (6)$$

Furthermore, let $\hat{x} - x$ be greater than a minimum distance x^* and

$$\frac{\sigma^2}{\mu} > \frac{2\sqrt{2(1+r)} - 1}{(\sqrt{2(1+r)} - 1)^{\frac{3\tau+1}{\tau}} - \frac{1}{2}} \doteq C(r, \tau)$$

which is trivially satisfied as $\tau \rightarrow 0$. Then, $x^* < \sigma$ and

$$\frac{f(\tau, x)}{\partial \sigma} > 0$$

If $x > \hat{x}$:

$$f(\tau, x) \sim \tau - O\left(\tau^{5/2} e^{-(x-\hat{x})^2/(2\sigma^2\tau)}\right) \quad (7)$$

with

$$\frac{f(\tau, x)}{\partial \sigma} < 0$$

Finally, as $\tau \rightarrow \infty$,

$$\frac{f(\tau, x)}{\partial \sigma} < 0$$

The above corollary says that for short horizons, the effect of volatility on expected occupation time depends sharply on the state variable's position relative to the investment threshold. When the process begins below the threshold, $x < \hat{x}$, higher volatility increases the probability of an upward crossing within a short interval, thereby increasing the expected time spent in the active region. In this regime, volatility acts as a force that facilitates

transition into the investment state.⁶

By contrast, when the process begins above the threshold, $x > \hat{x}$, the same increase in volatility raises the likelihood of downward excursions below the threshold, reducing expected occupation time in the active region. Thus, over short horizons, volatility has a locally stabilizing effect around the threshold: it pushes occupation probabilities toward an interior value by increasing the likelihood of crossing in either direction.

Importantly, this short-run effect differs fundamentally from the long-horizon comparative static. Because log drift satisfies $\bar{\mu} = \mu - \frac{1}{2}\sigma^2$, higher volatility also lowers long-run growth. As a result, while volatility can temporarily accelerate entry into the active region when the state starts below the threshold, over sufficiently long horizons the drift effect dominates, reducing the total expected time spent above the threshold. This generates a horizon-dependent reversal in the role of uncertainty: volatility operates as a short-run crossing mechanism but as a long-run drag on sustained occupancy of the investment region.

I formalize the above results into the following testable hypothesis,

H2. If the distance to investment is “far”, high firm-level risk is associated with higher investment but if the distance is “close”, high firm-level risk is associated with lower investment.

As supported by the theory, Figure 1 shows that the relationship between investment and uncertainty strongly depends on the initial distance between x_0 and $\hat{x}$. In this figure, I plot the firm’s expected investment rate varying the initial point x_0 along the x -axis while holding the horizon fixed. For various levels of σ , the optimal investment threshold $\hat{x}$ is between 0.2 – 0.3. Thus, the figure plots situations in which the firm is already above the action region, $x_0 > \hat{x}$, i.e., there is no delay and the firm immediately invests. In the left plot, the drift μ is set to a constant value while on the right, μ is increasing in σ to account for the possibility that higher volatility is compensated by higher growth rates.

Importantly, if the initial distance to the investment threshold is “far” (values of x_0 close to zero), high uncertainty is associated with higher investment. As discussed, the reason is that high volatility increases the likelihood of low probability events. As a result, the firm is likely to reach the investment threshold sooner and adopt a larger fraction of investment projects over a fixed time interval. On the other hand, if the firm is initially close to or is already above the threshold, we see that high uncertainty is associated with

⁶The requirement that $\hat{x} - x$ must be greater than a minimum distance x^* excludes scenarios in which the firm is on the cusp of investing, such that the immediate change to the boundary $d\hat{x}/d\sigma$ dominates the other channels. The condition that $\sigma^2/\mu > C(r, \tau)$ ensures that the minimum distance x^* is “short”, and is a sufficient condition such that $x^* < \sigma$. When $0 < \hat{x} - x < x^*$, the sign is ambiguous and must be evaluated numerically.

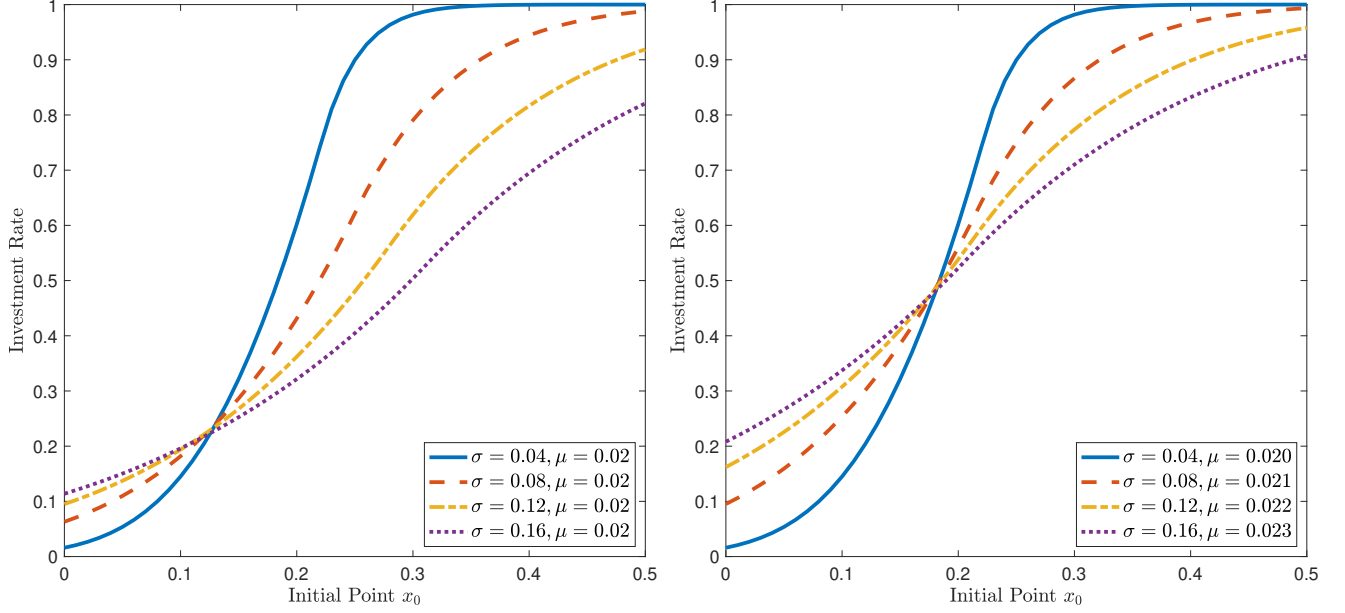


Figure 1: **Investment rate for different initial points x_0 over $T = 5$ years.** In the left panel, the optimal investment threshold $\hat{x}$ for each level of σ is: 0.22, 0.25, 0.28, 0.31 and for the right panel: 0.22, 0.22, 0.21, 0.21.

lower investment.⁷ The distance to investment $x_0 - \hat{x}$ therefore plays an important role in understanding the investment-uncertainty relationship.

With a final caveat, since the investment rate is also a function of the horizon T , Figure 2 plots the investment rate for different values of T while fixing $x_0 = 0$. On the left panel, μ is held fixed at $\mu = 0.02$ while on the right, μ is increasing in σ . Though the initial distance is far, the figure shows that extending the time interval at which we measure investment can reverse the investment-uncertainty relationship despite the preceding two figures. In this case, because $\mu > 0$, the stochastic variable x_t will eventually reach $\hat{x}$. By considering a long time interval, a higher value of σ needlessly increases the likelihood that projects fall out of profitability in the long-run. Hence, not only does the initial distance to investment matter, but so does the econometrician's choice of time interval.

Indeed, [Chang et al. 2024](#) finds a positive coefficient in regressions of physical capital investment on average uncertainty using quarterly data, whereas [Leahy and Whited 1996](#) and [Kermani and Ma 2023](#), who run the same regressions, find a negative coefficient using annual data. Using the tools studied in this section—the distance to investment and the choice of T —, we can rationalize these conflicting signs and explain why a positive relationship over short time intervals may turn to a negative relationship over longer ones.

⁷The fact that the sign change occurs before $\hat{x}$ supports Corollary 1, which requires a minimum distance for high firm-level risk to be associated with higher investment

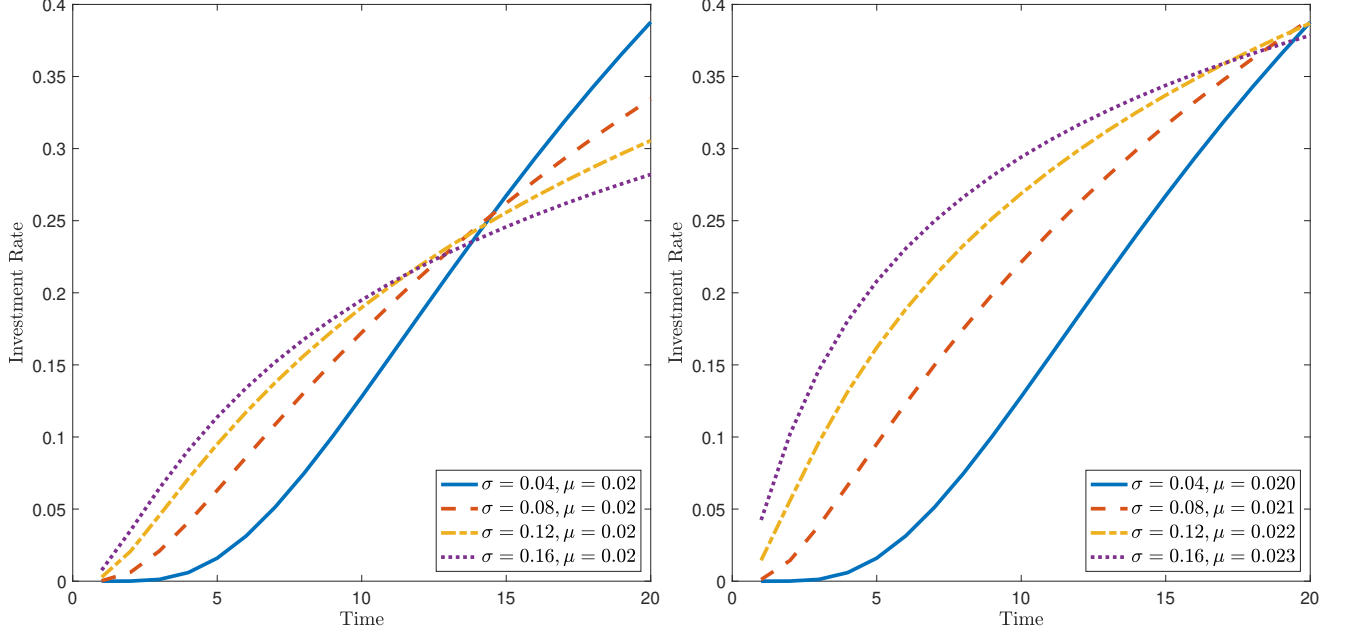


Figure 2: **Investment rate over different horizons T .**

2.3 Stochastic Volatility

The previous section focused on differences in the rate of investment if a firm faced different levels of constant volatility. In this section, I introduce stochastic volatility to study how the same firm responds to changes in uncertainty, separating a time-series effect from the cross-sectional effect of uncertainty. Now, suppose the economy alternates between two volatility regimes. I use $\eta^L dt, \eta^H dt$ to denote the risk-neutral probabilities of a transition. A H volatility regime can only be followed by a L regime and vice-versa. Furthermore, I assume there exists an exogenous stochastic discount factor in the economy, $dM_t/M_t = -r dt - \kappa dB_t$. The system of Hamilton-Jacobi-Bellman equations can be expressed (in terms of A),

$$\begin{aligned}
 rV^H(z, k, A) &= \zeta_0 A + (\mu - \kappa \sigma^H) A \frac{\partial V(z, k, A)}{\partial A} + \frac{1}{2} (\sigma^H)^2 A^2 \frac{\partial^2 V(z, k, A)}{\partial A^2} \\
 &\quad + \left[V^H(z + \tilde{\delta}, k + \tilde{k}, A) - V^H(z, k, A) - q \right] + \eta^L [V^L(z, k, A) - V^H(z, k, A)] \\
 rV^L(z, k, A) &= \zeta_0 A + (\mu - \kappa \sigma^L) A \frac{\partial V(z, k, A)}{\partial A} + \frac{1}{2} (\sigma^L)^2 A^2 \frac{\partial^2 V(z, k, A)}{\partial A^2} \\
 &\quad + \left[V^L(z + \tilde{\delta}, k + \tilde{k}, A) - V^L(z, k, A) - q \right] + \eta^H [V^H(z, k, A) - V^L(z, k, A)]
 \end{aligned}$$

with boundary conditions

$$\begin{aligned} V^H(z, k, A^{*H}) &= V^H(z + \mathfrak{z}, k + \mathfrak{k}, A^{*H}) - q \\ V^L(z, k, A^{*H}) &= V^L(z + \mathfrak{z}, k + \mathfrak{k}, A^{*H}) - q \end{aligned}$$

Applying the change of variable $x = \log A$ and defining the following:

$$\nu^H \doteq V_x^H \quad ; \quad \nu^L \doteq V_x^L$$

The ODE can be written as:

$$\nabla \begin{bmatrix} V^H \\ \nu^H \\ V^L \\ \nu^L \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{(1+r+\eta^L)}{.5(\sigma^H)^2} & -\frac{\mu-\kappa\sigma^H-.5(\sigma^H)^2}{.5(\sigma^H)^2} & -\frac{\eta^L}{.5(\sigma^H)^2} & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{\eta^H}{.5(\sigma^L)^2} & 0 & \frac{(1+r+\eta^H)}{.5(\sigma^L)^2} & -\frac{\mu-\kappa\sigma^L-.5(\sigma^L)^2}{.5(\sigma^L)^2} \end{bmatrix}}_{\doteq M} \begin{bmatrix} V^H \\ \nu^H \\ V^L \\ \nu^L \end{bmatrix} + \begin{bmatrix} 0 \\ \left[\zeta_0 + \frac{\tilde{\zeta}_1}{\sigma^H}\right] e^s - q \\ 0 \\ \left[\zeta_0 + \frac{\tilde{\zeta}_1}{\sigma^L}\right] e^s - q \end{bmatrix} \quad (8)$$

In Appendix 5.3, I show that there are constants a_1, a_3, b_1, b_3 , eigenvalues ω_1, ω_3 and elements of eigenvectors $\mathbf{v}^{(1)}, \mathbf{v}^{(3)}$ corresponding to these eigenvalues such that,

Proposition 2. In volatility regime H , investment occurs the first time

$$\tau^{H*} \doteq \inf \left\{ t : x_t \geq \log \left(\frac{\omega^{(1)}}{\omega^{(1)} - 1} \frac{v_1^{(3)} - v_3^{(3)}}{v_1^{(3)}(b_3 - a_3) - v_3^{(3)}(b_1 - a_1)} \times q \right) = \hat{x}^H \right\} \quad (9)$$

In volatility regime L , investment occurs the first time

$$\tau^{L*} \doteq \inf \left\{ t : x_t \geq \log \left(\frac{\omega^{(3)}}{\omega^{(3)} - 1} \frac{v_1^{(1)} - v_3^{(1)}}{v_1^{(1)}(b_3 - a_3) - v_3^{(1)}(b_1 - a_1)} \times q \right) = \hat{x}^L \right\} \quad (10)$$

Proof. See Appendix 5.3. □

Again, the optimal investment threshold is a ratio of the fixed investment cost divided by the marginal benefit to output of investing in this project, scaled by some constant. This time, the marginal benefit is a weighted average of two terms, splitting it into the expected marginal benefit at each regime.

Proposition 2 indicates that there are two investment thresholds ($\hat{x}^H$ and $\hat{x}^L$) corresponding to each regime. As a result, in order to compute an investment rate as before, it is necessary to account for the possibility of a regime change in the future.

More formally, let $f^H(t, x) \doteq \mathbb{E}_t \left[\int_t^T \mathbb{1}_{x_s \geq \hat{x}^H} \mathbf{d}s \right]$ and $f^L(t, x) \doteq \mathbb{E}_t \left[\int_t^T \mathbb{1}_{x_s \geq \hat{x}^L} \mathbf{d}s \right]$ denote the expected occupation time above the investment thresholds $\{\hat{x}^H, \hat{x}^L\}$. We can represent these conditional expectations as solutions to the equations

Lemma 3. $f^H(t, x)$ and $f^L(t, x)$ obey the following system of partial differential equations:

$$\begin{aligned} \frac{\partial f^H(t, x)}{\partial t} + \left[\mu^H - \frac{1}{2}(\sigma^H)^2 \right] \frac{\partial f^H(t, x)}{\partial x} + \frac{1}{2}(\sigma^H)^2 \frac{\partial^2 f^H(t, x)}{\partial x^2} + \eta^L [f^L(t, x) - f^H(t, x)] &= -\mathbb{1}_{x > \hat{x}^H} \\ \frac{\partial f^L(t, x)}{\partial t} + \left[\mu^L - \frac{1}{2}(\sigma^L)^2 \right] \frac{\partial f^L(t, x)}{\partial x} + \frac{1}{2}(\sigma^L)^2 \frac{\partial^2 f^L(t, x)}{\partial x^2} + \eta^H [f^H(t, x) - f^L(t, x)] &= -\mathbb{1}_{x > \hat{x}^L} \end{aligned}$$

with the terminal conditions $f^H(T, x) = 0$ and $f^L(T, x) = 0$.

Proof. See Appendix 5.4. □

Like before in Section 2.2, we proceed by reducing the above system of PDEs into a system of ODEs via the Laplace transform, then invert the solution to the system of ODEs to get back the expected occupation time from t to T .

Proposition 3. The expected investment rate in each regime is given by

$$\begin{aligned} I_T^H &= \frac{1}{T-t} \sum_j^2 \text{Res} \left(e^{\xi t} \bar{f}^H(\xi, x); \xi_j \right) \\ I_T^L &= \frac{1}{T-t} \sum_j^2 \text{Res} \left(e^{\xi t} \bar{f}^L(\xi, x); \xi_j \right) \end{aligned}$$

where $\xi_1 = 0$ and $\xi_2 = \eta^L + \eta^H$. The notation $\text{Res}(\cdot)$ indicates the residue of the function at the j th order pole ξ_j .⁸

Proof. See Appendix 5.3. □

⁸The definition of residues and poles is first presented in Appendix 5.2. To show the usefulness of the residue theorem, consider the integral

$$\int_{-\infty}^{\infty} \frac{1}{x^2 + 1} \mathbf{d}x = \pi$$

over the function $f(x) = \frac{1}{x^2 + 1}$. This function has two simple poles: i and $-i$. Along the complex plane, note that the integral $\int_{-\infty}^{\infty} f(x) \mathbf{d}x$ is a straight horizontal line along the real axis. To use contour integration (and hence, the residue theorem), we can close the contour arbitrarily along the upper-half plane, enclosing i . Then, by the residue theorem,

$$\oint_{\Omega} f(x) \mathbf{d}x = 2\pi i \text{Res}(f; i) = 2\pi i \times \lim_{x \rightarrow i} (x - i) \frac{1}{x^2 + 1} = 2\pi i \times \lim_{x \rightarrow i} \frac{1}{x + i} = 2\pi i \times \frac{1}{2i} = \pi$$

Using the above results, Figure 3 plots the expected investment rate in each volatility regime as a function of the initial point x_0 . Importantly, setting $\mu^H = \mu^L$, the figure shows that the firm has a lower investment rate in the higher volatility regime, regardless of the distance to investment. Hence, in most cases, an uncertainty shock leads to a decrease in investment.

Of course, the mechanism with stochastic volatility differs than in the static setting. The argument is as follows. Suppose the firm is in the high regime. With the specification that $\eta^L > \eta^H$, one risk of investing too early is that due to the high uncertainty, the level of the stochastic variable may reach an undesirable level before the arrival of the low regime, in which the firm is likely to be stuck around that range. Hence, the firm is reluctant to invest while in the high regime. We can see this more clearly in Figure 5. As η^L increases, as it becomes more likely that the low volatility regime will arrive, there is less risk of a sudden, large shock that makes the firm regret investing early. By shortening the expected duration of the high uncertainty regime, the firm is more willing to invest in the high regime.

Figure 4 then studies in setting in which μ^L and μ^H may differ. Importantly, the only scenario in which an uncertainty shock leads to an increase in investment is when $\mu^H > \mu^L$ and the initial distance is far. We can see this in the bottom-most plot. Furthermore, we can see that $\hat{x}^H < \hat{x}^L$ as well, indicating that a higher growth rate μ^H in the high volatility regime mitigates the precautionary motive. When the initial distance is far, high volatility again enables the firm to reach the target threshold sooner.

This result is consistent with [Atanassov et al. 2024](#) who find an increase in investment in response to uncertainty shocks. In their paper, the authors focus on R&D investment and argue that the effect of uncertainty may differ for R&D, in which projects typically take much longer to mature and turn profitable. Moreover, due to the nature of R&D, it is plausible that higher variance may be accompanied by a higher mean.

To summarize the results in Sections 2.2 and 2.3, I first studied the case in which the firm faces a constant level of uncertainty. By varying the value of σ , I showed that whether investment has a positive or negative relationship with uncertainty depends on the initial distance before which a project turns profitable. If the distance is far, high volatility increases the likelihood that the project turns profitable sooner while if the distance is close, higher volatility increase the likelihood that project becomes unprofitable.

In the time-series, firms typically reduce investment in response to uncertainty shocks. This is the standard result in the literature and in my paper, comes from a precautionary motive. On the other hand, I show that an exception occurs when uncertainty shocks are accompanied by increases in expected growth rates and at the same time, the initial distance is far.

In this partial-equilibrium setting, I provide a unified explanation for the mixed empirical findings on uncertainty and investment. The distance to investment is a simple statistic that

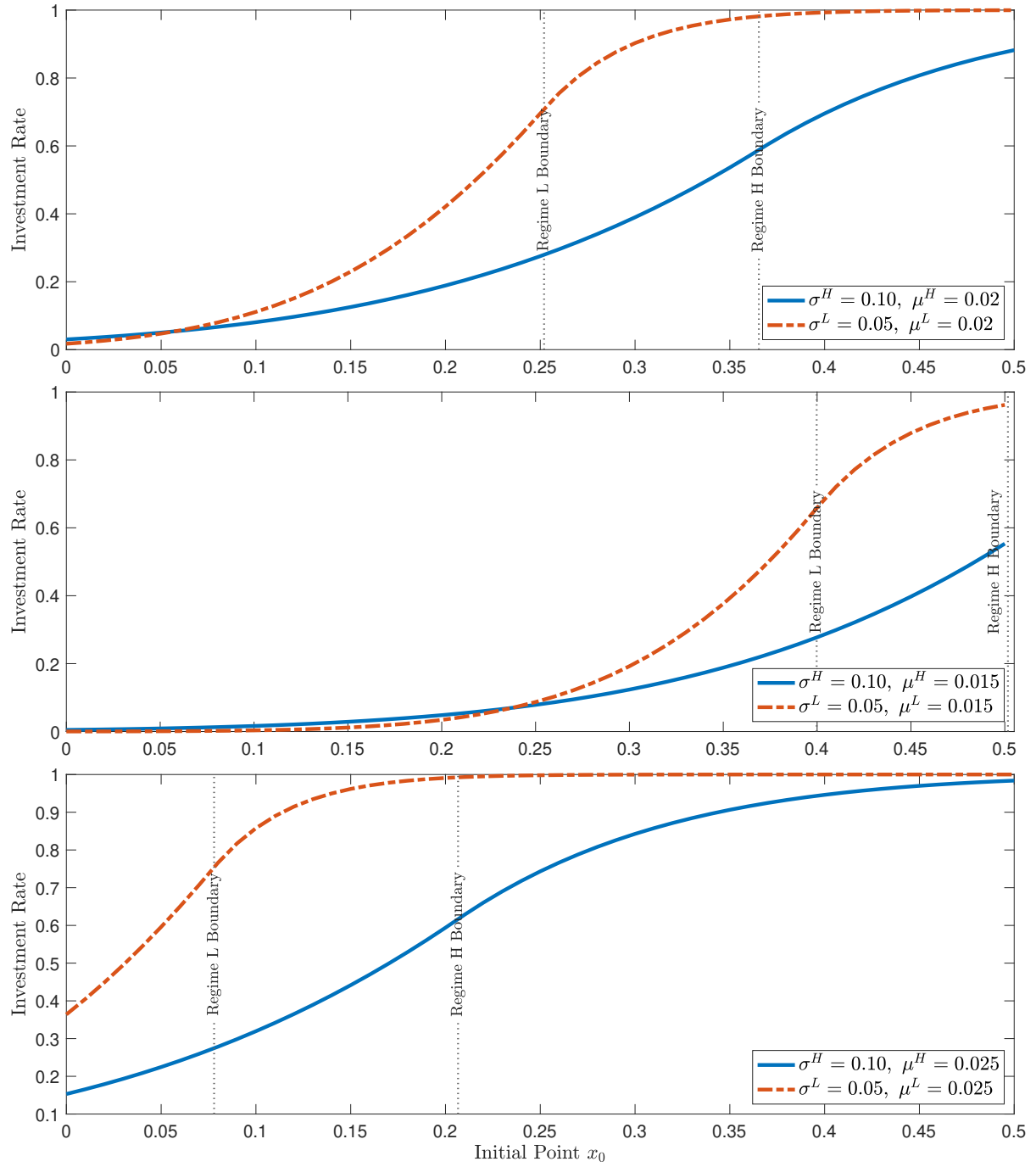


Figure 3: **Investment rate with stochastic volatility as a function of the initial point x_0 , fixing $T = 5$.** This figure plots the average investment rate over 5 years for different initial values of x_0 . In contrast to the next figure, $\mu^H = \mu^L$ in all cases.

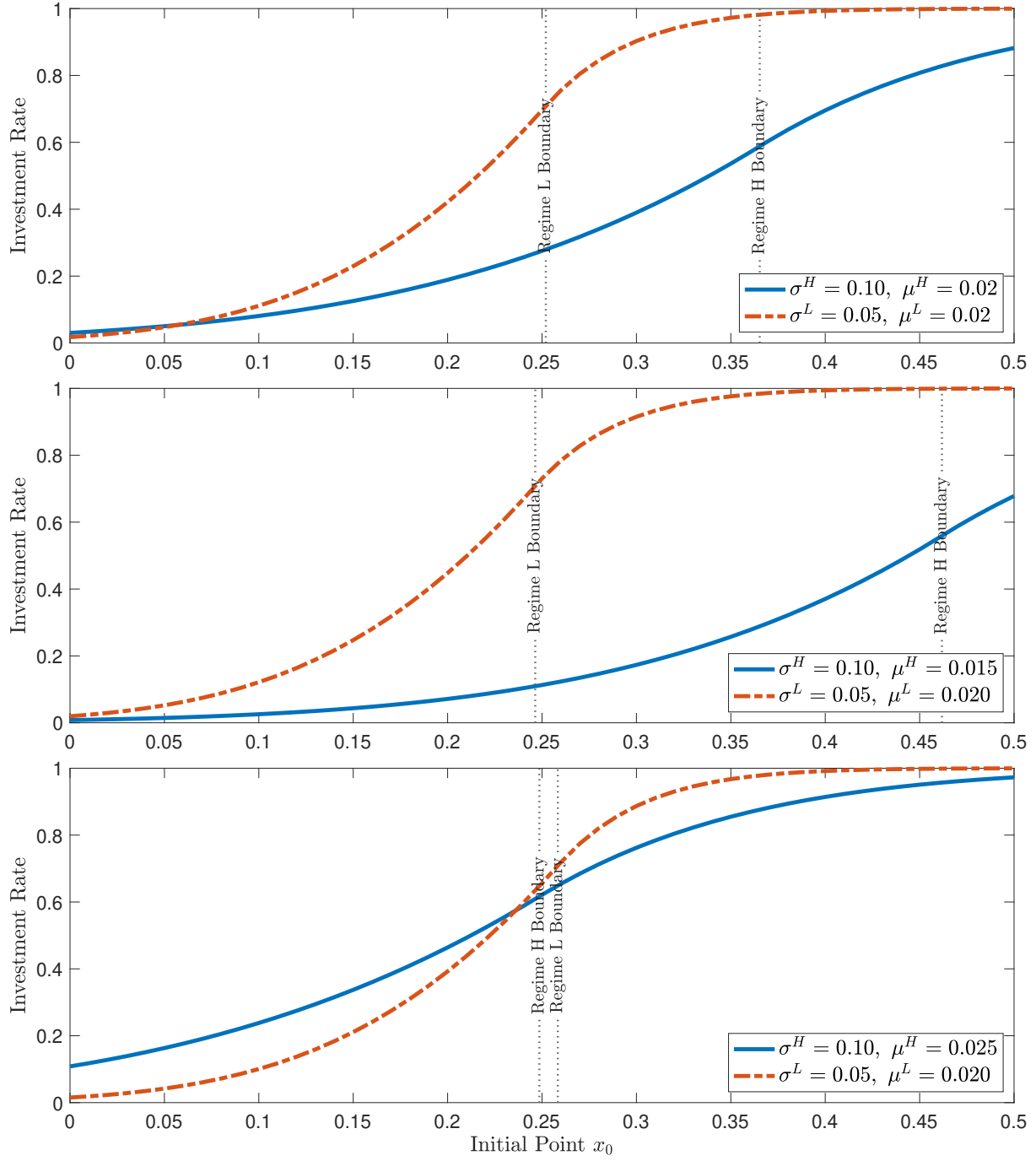


Figure 4: **Investment rate with stochastic volatility as a function of the initial point x_0 , fixing $T = 5$.** This figure plots the average investment rate over 5 years for different initial values of x_0 . Importantly, it separates investment into the following cases: $\mu^H = \mu^L$, $\mu^H > \mu^L$, and $\mu^H < \mu^L$.

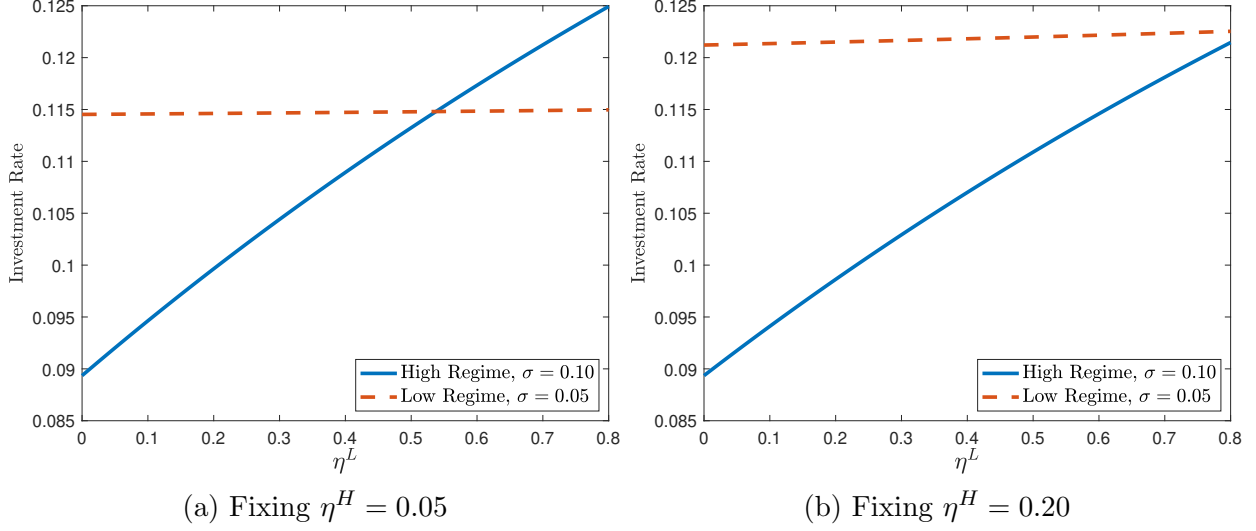


Figure 5: **Investment rate with stochastic volatility as a function of the arrival rate of the low regime η^L .** This figure shows the investment probability strictly increasing in the arrival rate of the low volatility regime while holding T fixed at 5 years and $\mu^H = \mu^L$.

organizes both time-series and cross-sectional variation, delivering clear conditions under which uncertainty increases or reduces investment.

2.4 The Consumer-Worker-Investor

This section addresses the concern that firms' cost of capital change in response to uncertainty shocks. I first present the setup of the model in Section 2.4.1 and derive results under static volatility, avoiding analysis since the intuition from the previous sections carry over. In Section 2.4.2, I move on to stochastic volatility.

2.4.1 Preferences and Technology

Suppose there exists a representative consumer-worker-investor who maximizes

$$\max_{c_s, \mathbf{d}l_s} \mathbb{E}_t \left[\int_t^\infty e^{-\rho(s-t)} \frac{c_s^{1-\gamma} - 1}{1-\gamma} \mathbf{d}s - \int_t^\infty e^{-\rho(s-t)} \vartheta_s \mathbf{d}l_s \right]$$

where the time rate of preference is ρ , risk aversion is γ and consumption is c_t . Importantly, whereas the previous section used q as an fixed investment cost in units of output, ϑ is disutility from capital installation, which requires labor. I assume that labor is not required in the production of the good but only in the installation of new capital when a firm adopts a project.⁹

⁹This assumption prevents lumpiness in consumption whenever a mass of firms adopt a project.

Furthermore, suppose there is a continuum of firms in unit measure indexed by i . Each firm produces a consumption good following the output process $y_t^i = \zeta_{n,t}^i A_t^i$ where $\zeta_{n,t}^i \doteq [\alpha(z_0^i + \sum_n \tilde{z}_n)^{\varepsilon} + (1 - \alpha)(k_0^i + \sum_n \tilde{k}_n)^{\varepsilon}]^{1/\varepsilon}$. To explain notation, (z_0^i, k_0^i) denotes firm i 's initial capital stock at date 0, and $(\tilde{z}_n, \tilde{k}_n)$ denotes capital accrued from the n th project completed by time t . Moreover, assume each firm is driven by an independent Brownian motion,

$$dA^i/A^i = \mu dt + \sigma^i \left(\sqrt{\varrho} dZ_t^* + \sqrt{1 - \varrho} dZ_t^i \right)$$

where dZ_t^* is aggregate risk and dZ_t^i is firm-specific risk. For simplicity, assume μ and ϱ are common across all firms and $dZ_t^i dZ_t^j = 0 \ \forall i \neq j$. However, σ^i is drawn independently from a Uniform distribution on $[\sigma_*, \sigma^*]$.

As before, investment projects arrive exogenously at rate λdt , independently across firms. By the law of large numbers, there is a mass of λ firms who receive a project at a given point in time. Moreover, with probability 1, there is an upper-limit of λt projects that are adopted by time t .

Since the representative consumer-worker-investor consumes all output, $c_t = y_t \doteq \int_0^1 y^i di$, the below lemma characterizes expected consumption after applying the change of variable, $x = \ln A$.

Lemma 4. Expected consumption at t is

$$\mathbb{E}_t[y_t] = e^{x_0 + \mu t} \times \frac{1}{\sigma^* - \sigma_*} \int_{\sigma_*}^{\sigma^*} \sum_{n=0}^{\infty} \zeta_n \overbrace{\frac{\Lambda_n(t, \sigma)^n e^{-\Lambda_n(t, \sigma)}}{n!}}^{= \Pr(N_t^i = n | \sigma^i = \sigma)} d\sigma \quad (11)$$

where

$$\Lambda_n(t, \sigma) = \lambda \int_0^t \Pr(\tau_n < s \mid \sigma) ds$$

and

$$\begin{aligned} \Pr(\tau_n < s \mid \sigma) &= \Phi\left(\frac{(\mu - .5\sigma^2) - (x_n^* - x_0)}{\sigma\sqrt{s}}\right) \\ &\quad + \exp\left(\frac{(\mu - .5\sigma^2)(x_n^* - x_0)}{\sigma^2}\right) \Phi\left(\frac{-(\mu - .5\sigma^2) - (x_n^* - x_0)}{\sigma\sqrt{s}}\right) \end{aligned}$$

and $f_{x_n^*}$ is the density function for the distribution of stopping times x^* for the n th project and $\Phi(\cdot)$ is the normal cumulative distribution function for the distribution of first passage time to x^* . Furthermore, define $w_t^i \doteq y_t^i/y_t$. The stochastic differential equation for the log

of aggregate output $X_t \doteq \log y_t$ is given by

$$dX_t = \left(\mu - \frac{1}{2} \varrho \bar{\sigma}_t^2 \right) dt + \bar{\sigma}_t \sqrt{\varrho} dZ_t^* + dJ_t$$

where $\bar{\sigma}_t = \int_0^1 w_t^i \sigma^i$ and

$$dJ_t = \log \left[1 + \int_0^1 w_t^i \left(\frac{\zeta_{n+1}^i}{\zeta_n^i} - 1 \right) dN_t^i di \right]$$

To complete the formulation of firm i 's problem, it is important to specify ϑ_t , the disutility from labor required to install new capital. I follow [Gârleanu et al. 2012](#) who consider a similar set-up in which a firm solves an optimal stopping problem to “plant” a new Lucas tree after the arrival of a new technological epoch. In their model, the representative consumer-worker-investor provides competitive labor to plant a tree, and as the owner of all firms, both pays and receives wage payments. The authors specify wages and the disutility from labor to match some stylized facts in the data. Namely, labor income must be cointegrated with total output. Moreover, by the envelope condition in dynamic programming, wage payments q_t must be given by $q_t = \frac{\vartheta_t}{U_c}$ where U_c is marginal utility of consumption. Further requiring that successive trees must be more costly to plant, the authors specify a functional form for ϑ_t . Adapting their approach to this setting, I set

$$\vartheta_t = \vartheta U_c y_t^{\gamma - (1-\gamma)/\vartheta} \quad (12)$$

Using $V(x^i, \zeta_n^i)$ to denote firm i 's value function (the subscript n denotes the number of projects completed by t , $\zeta_{n,t}^i \doteq [\alpha(z_0^i + \sum_n \tilde{z}_n)^\varepsilon + (1-\alpha)(k_0^i + \sum_n \tilde{k}_n)^\varepsilon]^{1/\varepsilon}$), the HJB equation in the no-action region between stopping times τ_n and τ_{n+1} can be written,

$$\begin{aligned} \rho V(x^i, \zeta_n^i) &= y^{-\gamma} \zeta_n^i e^{x^i} + \frac{\partial V(x^i, \zeta_n^i)}{\partial x^i} \left(\mu - \frac{1}{2} (\sigma^i)^2 \right) + \frac{1}{2} \frac{\partial^2 V(x^i, \zeta_n^i)}{\partial (x^i)^2} (\sigma^i)^2 \\ &\quad + \lambda \left(V(x^i, \zeta_{n+1}^i) - V(x^i, \zeta_n^i) \right) \end{aligned}$$

subject to the boundary conditions

$$\begin{aligned} V(x^i, \zeta_n^i) &= V(x^i, \zeta_{n+1}^i) - \vartheta y^{(\gamma-1)/\vartheta} \\ \frac{\partial V(x^i, \zeta_n^i)}{\partial x^i} &= \frac{\partial V(x^i, \zeta_{n+1}^i)}{\partial x^i} \end{aligned}$$

The following corollary describes the investment threshold for the $n+1$ th project.

Corollary 2. Let $\delta \doteq r - \mu + \frac{1}{2} (\sigma^i)^2$. The optimal stopping time for adoption of the $n+1$ th

project is the first time

$$\tau_{n+1}^* \doteq \inf \left\{ t : x_t \geq \log \left(\frac{\phi_1^+}{\phi_1^+ - 1} \frac{\vartheta y^{(\gamma-1)/\vartheta-1}}{\frac{\zeta_{n+1}^i - \zeta_n^i + \frac{\lambda}{\delta}(\tilde{\zeta}_{n+2}^i - \tilde{\zeta}_{n+1}^i)}{\delta + \lambda}} \right) = \hat{x} \right\} \quad (13)$$

where $\phi_1 \pm = -(\mu - .5(\sigma^i)^2)/(\sigma^i)^2 \pm \sqrt{2(\rho + \lambda)(\sigma^i)^2 + (\mu - .5(\sigma^i)^2)^2/(\sigma^i)^2}$

The equilibrium riskless rate is given by the drift of the consumer-worker-investor's marginal utility over consumption,

$$r_t = \rho + \gamma \left[\left(\mu - \frac{1}{2}(\gamma + 1)\varrho \bar{\sigma}_t^2 \right) + \Lambda(t, \bar{\sigma}) \times \Delta_t \right] \quad (14)$$

where

$$\Delta_t \doteq \frac{1}{\sigma^* - \sigma_*} \int_{\sigma_*}^{\sigma^*} \sum_{n=0}^{\infty} (\zeta_{n+1} - \zeta_n) \frac{\Lambda_n(t, \sigma)^n e^{-\Lambda_n(t, \sigma)}}{n!} d\sigma$$

The riskless rate is the sum of the following: the time rate of preference, a term proportional to the drift of output, a term related to prudence and precautionary demand, and a final term that captures the expected marginal increase in aggregate output over the next time interval due to the mass of firms that complete a new project.

2.4.2 Stochastic Volatility under General Equilibrium

I assume a regime change applies to all firms, and firm-level volatility in the high regime is a flat increase of σ_H^+ across all firms. First, the HJB equations can be written

$$\begin{aligned} \rho V^H(x^i, \zeta_n^i) &= y^{-\gamma} \zeta_n^i e^{x^i} + \frac{\partial V^H(x^i, \zeta_n^i)}{\partial x^i} \left(\mu - \frac{1}{2}(\sigma^{i,H})^2 \right) + \frac{1}{2} \frac{\partial^2 V(x^i, \zeta_n^i)}{\partial (x^i)^2} (\sigma^{i,H})^2 \\ &\quad + \lambda \left(V^H(x^i, \zeta_{n+1}^i) - V^H(x^i, \zeta_n^i) \right) + \eta^L \left(V^L(x^i, \zeta_n^i) - V^H(x^i, \zeta_n^i) \right) \end{aligned} \quad (15)$$

$$\begin{aligned} \rho V^L(x^i, \zeta_n^i) &= y^{-\gamma} \zeta_n^i e^{x^i} + \frac{\partial V^L(x^i, \zeta_n^i)}{\partial x^i} \left(\mu - \frac{1}{2}(\sigma^{i,L})^2 \right) + \frac{1}{2} \frac{\partial^2 V(x^i, \zeta_n^i)}{\partial (x^i)^2} (\sigma^{i,L})^2 \\ &\quad + \lambda \left(V^L(x^i, \zeta_{n+1}^i) - V^L(x^i, \zeta_n^i) \right) + \eta^H \left(V^H(x^i, \zeta_n^i) - V^L(x^i, \zeta_n^i) \right) \end{aligned} \quad (16)$$

subject to the boundary conditions

$$V^H(\widehat{x}^i, \zeta_n^i) = V^H(\widehat{x}^i, \zeta_{n+1}^i) - \vartheta y^{(\gamma-1)/\vartheta} \quad (17)$$

$$V^L(\check{x}^i, \zeta_n^i) = V^L(\check{x}^i, \zeta_{n+1}^i) - \vartheta y^{(\gamma-1)/\vartheta} \quad (18)$$

$$\frac{\partial V^H(\widehat{x}^i, \zeta_n^i)}{\partial x^i} = \frac{\partial V^H(\widehat{x}^i, \zeta_{n+1}^i)}{\partial x^i} \quad (19)$$

$$\frac{\partial V^L(\check{x}^i, \zeta_n^i)}{\partial x^i} = \frac{\partial V^L(\check{x}^i, \zeta_{n+1}^i)}{\partial x^i} \quad (20)$$

Dropping i superscripts, Apply the change of variable $x = \log A$ and defining the following:

$$\nu^H \doteq V_x^H \quad ; \quad \nu^L \doteq V_x^L$$

The ODE can be written as:

$$\nabla \begin{bmatrix} V^H \\ \nu^H \\ V^L \\ \nu^L \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{2(\rho+\lambda+\eta^L)}{(\sigma^H)^2} & -\frac{2(\mu-0.5(\sigma^H)^2)}{(\sigma^H)^2} & -\frac{\eta^L}{.5(\sigma^H)^2} & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{2\eta^H}{(\sigma^L)^2} & 0 & \frac{2(\rho+\lambda+\eta^H)}{(\sigma^L)^2} & -\frac{2(\mu-.5(\sigma^L)^2)}{(\sigma^L)^2} \end{bmatrix}}_{\doteq M} \begin{bmatrix} V^H \\ \nu^H \\ V^L \\ \nu^L \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{2}{(\sigma^H)^2} [y^{-\gamma} \zeta_n e^x + \lambda V^H(x, \zeta_{n+1})] \\ 0 \\ -\frac{2}{(\sigma^L)^2} [y^{-\gamma} \zeta_n e^x + \lambda V^L(x, \zeta_{n+1})] \end{bmatrix}$$

As before, the general solution to system of constant coefficient first order linear equations is

$$\mathbf{V} = \sum_i^4 c_i \mathbf{v}^{(i)} e^{\omega^{(i)} x}$$

where c_i are integration constants, $\mathbf{v}^{(i)}$ is the eigenvector associated with the i th eigenvalue $\omega^{(i)}$, and $\omega^{(i)}$ are the roots to $\det(M - \omega I)$.

More precisely,

$$\mathbf{v}^{(i)} = \begin{bmatrix} v_1^{(i)} \\ v_2^{(i)} \\ v_3^{(i)} \\ v_4^{(i)} \end{bmatrix} = \begin{bmatrix} 1 \\ \omega^{(i)} \\ \frac{2(\rho+\lambda+\eta^L) - (2\mu - (\sigma^H)^2)\omega^{(i)} - (\sigma^H \omega^{(i)})^2}{2\eta^L} \\ \omega^{(i)} v_3^{(i)} \end{bmatrix}$$

Further requiring that $e^{\omega^{(i)}x} \rightarrow 0$ as $x \rightarrow -\infty$, we have the two solutions:

$$\begin{aligned} V^H(x, \zeta_n^i) &= \frac{y^{-\gamma}}{\rho + \lambda - \mu} \left(\zeta_n + \frac{\lambda \zeta_{n+1}}{\rho - \mu} \right) e^x + c_1 e^{\omega^{(1)}x} + c_2 e^{\omega^{(2)}x} \\ V^L(x, \zeta_n^i) &= \frac{y^{-\gamma}}{\rho + \lambda - \mu} \left(\zeta_n + \frac{\lambda \zeta_{n+1}}{\rho - \mu} \right) e^x + c_1 v_3^{(1)} e^{\omega^{(1)}x} + c_2 v_3^{(2)} e^{\omega^{(2)}x} \end{aligned}$$

and the integration constants c_1, c_2 :

$$c_1 = \frac{e^{\omega^{(2)}\hat{x}} (\vartheta y^{(\gamma-1)/\vartheta} + \Gamma e^{\hat{x}}) - e^{\omega^{(2)}\tilde{x}} v_3^{(2)} (\vartheta y^{(\gamma-1)/\vartheta} + \Gamma e^{\tilde{x}})}{e^{\omega^{(1)}\hat{x} + \omega^{(2)}\tilde{x}} v_3^{(2)} - e^{\omega^{(2)}\hat{x} + \omega^{(1)}\tilde{x}} v_3^{(1)}} \quad (21)$$

$$c_2 = \frac{e^{\omega^{(1)}\tilde{x}} v_3^{(1)} (\vartheta y^{(\gamma-1)/\vartheta} + \Gamma e^{\tilde{x}}) - e^{\omega^{(1)}\hat{x}} (\vartheta y^{(\gamma-1)/\vartheta} + \Gamma e^{\hat{x}})}{e^{\omega^{(1)}\hat{x} + \omega^{(2)}\tilde{x}} v_3^{(2)} - e^{\omega^{(2)}\hat{x} + \omega^{(1)}\tilde{x}} v_3^{(1)}} \quad (22)$$

where

$$\Gamma = \frac{y^{-\gamma}(\zeta_n - \zeta_{n+1})}{\rho + \lambda - \mu} \quad (23)$$

and the optimal thresholds $\hat{x}, \tilde{x}$ are determined jointly by Equations (19) and (20),

$$\frac{y^{-\gamma}}{\rho + \lambda - \mu} \left(\zeta_n + \frac{\lambda \zeta_{n+1}}{\rho - \mu} \right) e^{\hat{x}} + c_1 \omega^{(1)} e^{\omega^{(1)}\hat{x}} + c_2 \omega^{(2)} e^{\omega^{(2)}\hat{x}} - \frac{y^{-\gamma} \zeta_{n+1}}{\rho - \mu} e^{\hat{x}} = 0 \quad (24)$$

$$\frac{y^{-\gamma}}{\rho + \lambda - \mu} \left(\zeta_n + \frac{\lambda \zeta_{n+1}}{\rho - \mu} \right) e^{\tilde{x}} + c_1 v_4^{(1)} e^{\omega^{(1)}\tilde{x}} + c_2 v_4^{(2)} e^{\omega^{(2)}\tilde{x}} - \frac{y^{-\gamma} \zeta_{n+1}}{\rho - \mu} e^{\tilde{x}} = 0 \quad (25)$$

The equilibrium riskless rate is given by

$$r_t = \rho + \gamma \left[\left(\mu - \frac{1}{2}(\gamma + 1) \varrho \bar{\sigma}_t^2 \right) + \Lambda(t, \bar{\sigma}) \times \Delta_t^{\{L, H\}} \right] \quad (26)$$

and $\Lambda(t, \bar{\sigma}) \times \Delta_t^{\{L, H\}}$ is the expected marginal increase in output over the next time interval given the current regime, which can be completed via simulation.

Putting the model ingredients together, Figure 7 documents a sharp and economically meaningful nonlinearity in the investment response to uncertainty shocks, conditioned on the prevailing state of aggregate output. The figure reports the cumulative difference in investment rates between the high-volatility regime (H) and the low-volatility counterfactual (L) over an 18-month horizon, for initial detrended output levels ranging from $y_0 = -0.50$ to $y_0 = +0.05$ via simulation. Two distinct regimes emerge cleanly. When the economy is sufficiently depressed — specifically, when detrended output lies below approximately -0.20 — the arrival of an uncertainty shock *stimulates* cumulative investment relative to the low-volatility counterfactual, with the effect peaking at +1.5 percentage points at $y_0 =$

Table 1: Calibration Parameters

<i>Parameter</i>	<i>Symbol</i>	<i>Value</i>
<i>Panel A: Preferences and Technology</i>		
Discount rate	ρ	0.01
Risk aversion	γ	3.0
Factor loading on aggregate risk	φ	0.6
Trend growth rate	μ	0.015
Installation cost elasticity	ϑ	3.0
<i>Panel B: Firm Heterogeneity</i>		
Idiosyncratic volatility, lower bound	$\underline{\sigma}$	0.075
Idiosyncratic volatility, upper bound	$\bar{\sigma}$	0.110
Volatility increment in high regime	σ_H^+	0.08
Std. of log initial capital stock	σ_ζ	0.30
<i>Panel C: Regime-Switching Process</i>		
Transition rate, high $\rightarrow$ low	η_{HL}	3.0
Transition rate, low $\rightarrow$ high	η_{LH}	0.30
Mean duration of high regime (months)		4.0
Unconditional probability of high regime		0.091
<i>Panel D: Investment Technology</i>		
Project arrival rate (per year)	λ	1.0
<i>Panel E: Simulation Design</i>		
Number of firms per simulation	N	500
Number of simulation pairs	R	500
Time step (years)	dt	1/52
Event-study horizon (years)	T_{post}	1.5

-0.26 . Conversely, when output is near or above trend, the same uncertainty shock *depresses* investment by between 0.5 and 1.60 percentage points, with the largest contraction occurring at $y_0 = -0.14$. The sign reversal is sharp, statistically robust given the tight confidence intervals in the recession states, and monotone across the state space once it occurs.

Figure 6 summarizes the mechanism underlying this nonlinearity. For depressed states (green lines), the $H - L$ gap is negligible or slightly negative on impact but turns persistently positive within the first quarter and continues widening throughout the 18-month window, reaching $+0.40$ to $+0.70$ percentage points by month 18. This pattern is inconsistent with a simple wait-and-see story, which would predict an initial depression followed by a strong rebound due to pent-up demand. Instead, we observe the same mechanism—the distance to

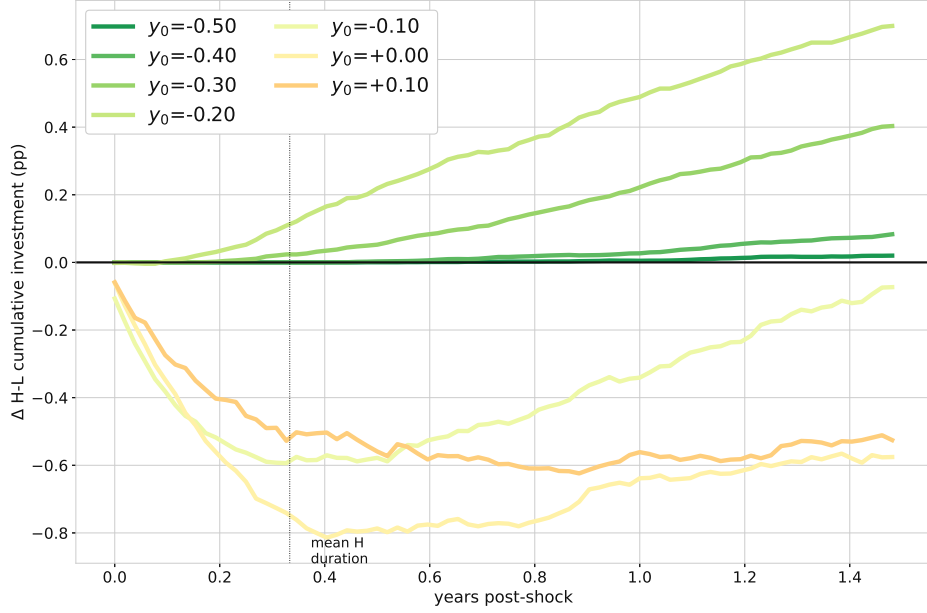


Figure 6: Difference in Cumulative Investment Response to Uncertainty Shock — Time Series by Initial State. Each line plots the mean cumulative difference in investment rates between the high-volatility regime (H) and the low-volatility counterfactual (L) over a 18-month horizon following the arrival of an uncertainty shock, conditional on the economy’s initial detrended output level $y_0 = \log y_t - \mu t$. Simulations are initialized with the cross-sectional distribution of firm-level log-productivity drawn from the ergodic distribution centered at y_0 , with both arms sharing identical initial firm positions and Poisson project-arrival timing, differing only in the volatility regime at $t = 0$. The high-volatility regime raises idiosyncratic firm volatility by $\sigma_H^+ = 0.08$ and reverts stochastically to the low-volatility regime with a mean H duration of four months. Lines shaded green denote initial states sufficiently below trend ($y_0 \leq -0.30$); lines shaded yellow-to-orange denote states near or above trend ($y_0 \geq -0.20$).

investment—at play: in a recession, the cross-sectional distribution of firm-level productivity is far below the investment threshold, and the wider diffusion induced by H systematically shifts more firms into the investment region. For near-trend and above-trend states (yellow and orange lines), the pattern reverses sharply on impact — the $H - L$ gap falls to between -0.50 and -0.80 percentage points within the first two months, reaching a trough near the mean H duration of four months. This on-impact depression reflects the *fallback channel*: when aggregate output is near trend, a non-trivial mass of firms already sits above the investment threshold awaiting a Poisson arrival, and the higher volatility in H increases the probability that these firms diffuse back below the threshold before the arrival occurs, forestalling investment, consistent with Corollary 1. The subsequent partial recovery visible for the $y_0 = -0.10$ lines reflects a post- H rebound as pent-up investment materializes once the regime reverts.

These results carry direct implications for the state-dependence of the effect of uncertainty shocks. The conventional view, originating with [Bernanke 1983](#) and formalized in [Dixit and Pindyck 1994](#), treats uncertainty as uniformly contractionary for investment through the

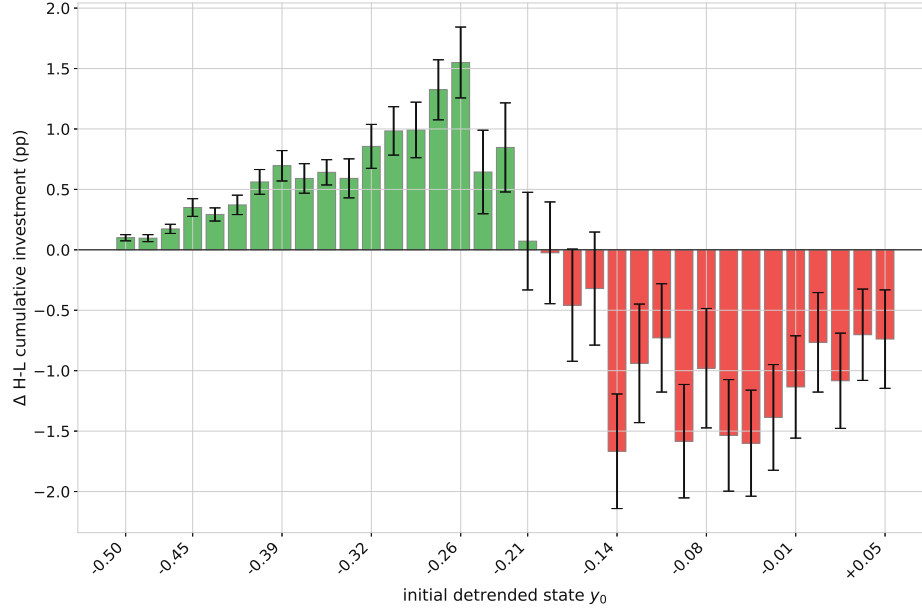


Figure 7: **Terminal $H - L$ Investment Gap by Initial Aggregate State.** Bars report the mean cumulative $H - L$ investment differential (in percentage points of firms investing) measured at the 18-month horizon, as a function of the initial detrended output level y_0 . Green bars indicate states where the uncertainty shock stimulates cumulative investment relative to the low-volatility counterfactual; red bars indicate states where the shock depresses it. Error bars denote 95-percent bootstrap confidence intervals constructed from 500 bootstrap draws, resampling with replacement from the set of simulation pairs at each initial state.

option value of waiting. This general equilibrium model and simulations qualify the mainstream view in a quantitatively important way: the contractionary effect is concentrated in economies operating near or above potential, where sufficient productive capacity is already poised to be deployed. In depressed economies, where firms are far from their investment thresholds, uncertainty can be *expansionary* by facilitating threshold crossings that would not otherwise occur under low volatility. This state-dependence has direct policy relevance. Countercyclical uncertainty — the well-documented empirical regularity that measures of economic uncertainty spike precisely during recessions [Bloom 2009](#) and [Jurado et al. 2015](#) — may therefore be less damaging to investment than aggregate estimates suggest, since recessions are exactly the states where the model predicts uncertainty stimulates rather than depresses capital formation. The results also suggest that the investment response to a given uncertainty shock depends critically on whether it arrives during a contraction or an expansion, a heterogeneity that standard linear impulse-response analysis systematically obscures.

3 Empirical Analysis

I restate the two testable predictions,

H1. The persistence of a firm’s investment rate is decreasing in its squared Sharpe ratio.

H2. If the distance to investment is “far”, high firm-level risk is associated with higher investment but if the distance is “close”, high firm-level risk is associated with lower investment.

3.1 Data & Methodology

To test the two predictions, we require estimates of firms’ investment rate persistence, their Sharpe ratios, and distance to investment. I address each in turn.

3.1.1 Investment Persistence & Bootstrap

Treating persistence as a time-invariant parameter, I estimate ρ via a simple AR(1) specification:

$$\text{Investment rate}_{i,t+1} = \alpha_i + \rho_i \text{Investment Rate}_{i,t} + \varepsilon_{i,t+1}$$

where the investment rate is computed as $\text{ik}_{i,t+1} = \text{capx}_{t+1} / (.5\text{ppent}_t + .5\text{ppent}_{t+1})$ following [Belo et al. 2014](#). The estimates of firm-level persistence $\hat{\rho}_i$ is then used in the second-stage Fama-Macbeth regressions,

$$\hat{\rho}_i = \mathbf{x}_{i,t}\boldsymbol{\xi} + \beta_t \text{SR}_{i,t}^2 + \gamma_j + \varepsilon_{i,t} \quad \text{for each } t$$

where $\mathbf{x}_{i,t}$ is a vector of firm controls, $\text{SR}_{i,t}^2$ is an estimate of the firm’s Sharpe ratio, and γ_j are industry fixed effects. I then find the average of the cross-sectional estimates

$$\hat{\beta} = \frac{1}{T} \sum_{t=1}^T \hat{\beta}_t \quad ; \quad \sigma^2(\hat{\beta}) = \frac{1}{T(T-1)} \sum_{t=1}^T (\hat{\beta}_t - \hat{\beta})^2$$

The main concern is that $\hat{\rho}_i$ is estimated in the data. This is problematic since firms that appear in only finite samples may have attenuated estimates of $\hat{\rho}$ given the idiosyncratic shocks at that point in time; since these are individual firm-level regressions, firm, industry or time fixed-effects cannot be included to account for unobserved heterogeneity.

Thus, to account for uncertainty in the first-stage estimates of investment persistence, I employ a parametric residual bootstrap. In the first stage, I estimate firm-level AR(1) regressions once on the original data, yielding the estimates $\hat{\rho}_i$, $\hat{\alpha}_i$, and centred residuals $\hat{\varepsilon}_{i,t} = \text{IR}_{i,t} - \hat{\alpha}_i - \hat{\rho}_i \text{IR}_{i,t-1}$ where IR stands for Investment Rate. In each of $B = 500$

replications, I then draw N firms with replacement. For each resampled firm, I construct a synthetic investment rate series by drawing residuals with replacement from that firm's own empirical residual distribution and recursively rebuild the series as $\widetilde{\text{IR}}_{i,t} = \widehat{\alpha}_i + \widehat{\rho}_i \widetilde{\text{IR}}_{i,t-1} + \varepsilon_{i,t}^*$, where $\varepsilon_{i,t}^* \stackrel{\text{iid}}{\sim} \widehat{F}_i$ and $\widehat{F}_i$ is the empirical distribution of firm i 's residuals. This addresses the concern that had the firm been subject to a different sequence of shocks, the estimate of $\widehat{\rho}$ would have changed. I then re-estimate persistence on the synthetic series, yielding $\widehat{\rho}_i^{(b)}$. Next, I recompute the within-year Sharpe ratio ranks on the resampled cross-section. The second-stage cross-sectional regression is run at each t using $\widehat{\rho}_i^{(b)}$ as the dependent variable, and coefficients are averaged across years to yield $\widehat{\beta}^{(b)}$. The bootstrap standard error is the standard deviation of $\{\widehat{\beta}^{(b)}\}_{b=1}^B$.

The procedure is formalized as follows. The first stage is estimated once on the original data as before, with centered residuals

$$\widehat{\varepsilon}_{i,t} \leftarrow \widehat{\varepsilon}_{i,t} - \widehat{\varepsilon}_i. \quad (27)$$

For each replication $b = 1, \dots, B$:

Step 1: $\mathcal{S}^{(b)}$ = draw N firms with replacement from $\{1, \dots, N\}$,

Step 2: $\varepsilon_{i,t}^* \stackrel{\text{iid}}{\sim} \widehat{F}_i \quad \forall i \in \mathcal{S}^{(b)}$,

Step 3: $\widetilde{\text{IR}}_{i,t} = \widehat{\alpha}_i + \widehat{\rho}_i \widetilde{\text{IR}}_{i,t-1} + \varepsilon_{i,t}^*$,

Step 4: $\widehat{\rho}_i^{(b)} = \text{AR}(1)(\widetilde{\text{IR}}_{i,\cdot})$,

Step 5: $\widetilde{\text{SR}}_{i,t}^{(b)} = \text{Rank}(\text{SR}_{i,t}^2 \mid i \in \mathcal{S}^{(b)}, t) \quad \forall t$,

Step 6: $\widehat{\beta}_t^{(b)} = \text{CS-reg}(\widehat{\rho}_i^{(b)}, \widetilde{\text{SR}}_{i,t}^{(b)}, \mathbf{x}_{i,t} \mid i \in \mathcal{S}^{(b)}) \quad \forall t$,

Step 7: $\widehat{\beta}^{(b)} = \frac{1}{T} \sum_{t=1}^T \widehat{\beta}_t^{(b)}$.

The bootstrap standard error and t -statistic are as in the main table. Two additional design choices are worth noting. First, residuals are resampled nonparametrically: no distributional assumption is imposed on $\widehat{F}_i$, so fat tails and skewness in firm-level investment shocks are preserved. Second, the original estimate $\widehat{\rho}_i$ is used to generate the synthetic series in Step 3 but re-estimated in Step 4. This ensures the data-generating process is centered at the original estimate while the re-estimation in Step 4 recovers the distribution of $\widehat{\rho}_i^{(b)}$ around $\widehat{\rho}_i$, which addresses the fact that $\widehat{\rho}_i$ is an estimated dependent variable.

3.1.2 Sharpe Ratio

In principle, the Sharpe ratio should correspond to the parameters μ and σ associated with the firm’s productivity process—which is unobserved. However in a production economy, the return on equity is equal to the drift of the stochastic process. I take a reduced-form approach, treating the firm’s expected unlevered return on equity via CAPM as a proxy for μ . For each firm, I run rolling daily regressions of firm excess returns on excess returns on the value-weighted CRSP index, from year $t - 4$ to t . Using the beta from that regression, I de-lever beta by $1 + D/E$ to find the unlevered beta, and use $r^f = 0.035$ and $R^M - r^f = 0.06$ as fixed constants. For robustness, I also use the expected return using the predicted values from daily regressions using the Fama-French 5 factor model, as well as historical daily returns.

For σ , I use 252 times the volatility of daily stock returns over the previous calendar year as is standard in the literature. All data is June-to-June of that fiscal year.

Importantly, as Table 2 shows, the distribution of the Sharpe ratio is highly skewed, especially when using historical stock returns as the estimate of μ . To address these concerns, I take two approaches: (1) within each year, firms are ranked by their estimated SR^2 , where a high rank corresponds to a high SR^2 . The variable Rank SR^2 is the firm’s ranking at time t . (2) I take logs of SR^2 . In both cases, we lose interpretability of the regression coefficients, but this address concerns over the bias due to the extensive right-tail of the distribution.

3.1.3 Distance to Investment

For each firm i in industry j and fiscal year t , I define a binary outcome variable indicating whether the firm’s investment rate falls in the top quartile of its industry-year distribution. Specifically, let pct_ik_{ijt} denote the percentile rank of firm i ’s investment rate within the set of firms sharing the same 2-digit SIC code and fiscal year. The outcome variable is then

$$D_{ijt} = \mathbb{1}[pct_ik_{ijt} \geq 0.75] , \quad (28)$$

where $\mathbb{1}[\cdot]$ is the indicator function, so that $D_{ijt} = 1$ for approximately the top 25% of firms by investment intensity within each industry-year cell.

I model the conditional probability of high investment via a probit specification:

$$\Pr(D_{ijt} = 1 \mid \mathbf{x}_{it}) = \Phi(\mathbf{x}_{it}'\boldsymbol{\beta}) , \quad (29)$$

where $\Phi(\cdot)$ is the standard normal CDF and $\mathbf{x}_{it}$ is a vector of firm-level predictors measured at the end of fiscal year $t - 1$. The predictor vector includes: the Whited-Wu index of financial constraints, operating cash flows scaled by assets, Tobin’s q , the market-to-book ratio, asset tangibility, payout ratio, return on assets, intangible intensity, firm size ($\log Sales$), cash

holdings, R&D intensity, EBIT margin, log total assets, capital expenditure intensity, and leverage. Prior to estimation, each continuous predictor is standardized using training-sample means and standard deviations:

$$\tilde{x}_{it}^{(k)} = \frac{x_{it}^{(k)} - \bar{x}_{\tau < t}^{(k)}}{\hat{\sigma}_{\tau < t}^{(k)}}, \quad (30)$$

where $\bar{x}_{\tau < t}^{(k)}$ and $\hat{\sigma}_{\tau < t}^{(k)}$ denote the mean and standard deviation of the k -th predictor computed over all training observations strictly prior to year t . This ensures that no information from the evaluation period contaminates feature construction.

To guard against look-ahead bias and provide a genuine out-of-sample evaluation, I estimate the model on an expanding window. For each year t , the parameter vector β_t is estimated by maximum likelihood using all available firm-year observations from years $\tau < t$:

$$\hat{\beta}_t = \arg \max_{\beta} \sum_{\tau < t} \sum_i \left[D_{i\tau} \log \Phi(\tilde{\mathbf{x}}'_{i\tau} \beta) + (1 - D_{i\tau}) \log(1 - \Phi(\tilde{\mathbf{x}}'_{i\tau} \beta)) \right], \quad (31)$$

subject to a minimum of five training years and at least 200 firm-year observations. Standard errors are clustered at the firm level to account for serial correlation in unobserved determinants of investment. The out-of-sample predicted probability for firm i in year t is then

$$\hat{p}_{it} = \Phi(\tilde{\mathbf{x}}'_{it} \hat{\beta}_t), \quad (32)$$

where $\hat{\beta}_t$ is estimated on data from years strictly prior to t . Finally, I measure distance as

$$\text{InvDist}_{it} = 1 - \hat{p}_{it}$$

I evaluate out-of-sample predictive performance using two complementary metrics. The first is the area under the receiver operating characteristic curve (AUC), which measures the probability that a randomly drawn high-investment firm receives a higher predicted probability than a randomly drawn low-investment firm:

$$\text{AUC}_t = \Pr(\hat{p}_{it} > \hat{p}_{i't} \mid D_{it} = 1, D_{i't} = 0). \quad (33)$$

A value of 0.5 corresponds to random classification, and a value of 1 indicates perfect discrimination. The second metric is the Brier score, defined as the mean squared error of the probabilistic forecasts:

$$\text{BS}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} (\hat{p}_{it} - D_{it})^2. \quad (34)$$

Lower values indicate better-calibrated forecasts. I benchmark the Brier score against a naive classifier that assigns the unconditional base rate $\bar{p} = \Pr(D_{ijt} = 1)$ to every firm, yielding a benchmark of $BS^{\text{naive}} = \bar{p}(1 - \bar{p})$.

Figure IA.11 in the Appendix shows the time-series AUC and Brier Score for the rolling probit regressions. The AUC is persistently over 80% with a one-time drop during Covid, while the Brier Score hovers below 20%. Thus, both these diagnostics support the claim that the out-of-sample probit is a valid predictor of next period’s investment.

Table 2: Descriptive Statistics

	Mean	Std. Dev.	p5	p25	p50	p75	p95
Investment Rate	0.23	0.14	0.04	0.11	0.19	0.32	0.50
Investment Rate Persistence : $\hat{\rho}$	0.41	0.29	-0.08	0.23	0.44	0.62	0.82
Distance : $1 - \Pr(\widehat{\mathbb{1}_{ik_{t+1}}})$	0.77	0.24	0.18	0.71	0.84	0.94	1.00
SR ² via CAPM	0.10	0.29	0.00	0.01	0.03	0.11	0.39
SR ² via historical data	1.06	2.15	0.00	0.11	0.45	1.14	4.13

3.2 Results

Table 3 provides direct evidence for one of the model’s sharpest structural predictions: that the rate at which a firm’s investment behaviour reverts to its long-run ergodic level is governed by the squared Sharpe ratio of its productivity process. The mechanism comes from the spectral decomposition of the expected occupation time, where the transient component decays exponentially at rate $\bar{\mu}^2/(2\sigma^2)$. A firm with a high signal-to-noise ratio in its productivity dynamics has a high squared Sharpe ratio, meaning that it quickly shakes off any temporary shocks and converges rapidly to its long-run ergodic level. A firm with a low Sharpe ratio, by contrast, is subject to persistent uncertainty about whether it will reach the investment threshold, and carries the informational content of its current state for many periods. The negative and highly significant coefficients on both Rank SR² and log SR² across all specifications confirm this prediction: firms ranked higher within their industry by squared Sharpe ratio exhibit lower investment rate persistence, consistent with the model.

However, the economic magnitude of the result is small. The coefficient of -8.26 on Rank SR² $\times 100$ in column (1) implies that a one standard-deviation increase in within-industry rank is associated with a .1 standard deviation decrease in $\hat{\rho}_i$. While modest in absolute terms, this effect may be influenced by attenuation bias inherent in the estimation procedure: both the dependent variable $\hat{\rho}_i$ and the independent variable SR²_{it} are estimated from rolling time-series windows with substantial sampling noise, and classical errors-in-variables theory predicts that the true structural relationship is larger than the

Table 3: This table is the time-averaged cross-sectional regression (Fama-Macbeth):

$$\hat{\rho}_i = \text{Investment rate persistence}_i = \mathbf{x}_{i,t}\boldsymbol{\xi} + \beta_t \text{SR}_{i,t}^2 + \gamma_j + \varepsilon_{i,t} \quad \text{for each } t$$

where $\hat{\rho}_i$ is estimated in first-stage time-series regressions at the firm-level. $\mathbf{x}_{i,t}$ is a vector of firm controls such as Tobin's Q , $\log(\text{Assets})$, $\log(\text{Sales})$, and Leverage. The main independent variable is SR^2 , the squared Sharpe ratio. To address the skewness in Sharpe ratio's at the firm-level, I transform the data in the following ways: (1) within each year, firms are ranked by their estimated SR^2 , where a high rank corresponds to a high SR^2 . The variable Rank SR^2 is the firm's ranking at time t . (2) I take logs of SR^2 . Two sets of test statistics are reported. The first are computed from the Fama-Macbeth procedure, while the second are bootstrapped standard errors that takes into account the fact that both persistence and SR^2 are estimated in the data. The detailed procedure is outlined in Appendix (6.1). To compute μ , I use the unlevered annual expected return via CAPM, which is equal to the expected return on assets, using beta calculated over rolling 5-year windows on daily data. The results are robust to the alternative choices such as the expected return from a Fama-French 5-factor model. Firm-level uncertainty is the annualized volatility of equity, as is standard in the literature. Data is 1984-2022.

Dependent Variable: Investment rate persistence _i				
	(1)	(2)	(3)	(4)
Rank $\text{SR}^2 \times 100$	-8.26*** (-5.89) (-7.49)		-3.21*** (-6.72) (-3.23)	
$\log \text{SR}^2 \times 100$		-0.86*** (-5.11) (-7.15)		-0.33*** (-4.86) (-3.02)
Firm controls	No		Yes	
Fixed effects	sic-2		sic-2	
Adjusted R^2	0.097	0.093	0.097	0.093
Observations	65,520	65,520	65,520	65,520
<i>t</i> statistics in parentheses				
* $p < 0.10$				
** $p < 0.05$				
*** $p < 0.01$				

observed coefficient. The fact that the bootstrapped standard errors—which account for the generated-regressor nature of both variables—remain highly significant across all specifications indicates that the attenuation bias is compressing the magnitude of the relationship but not its statistical reliability. The log specification in columns (2) and (4) provides a complementary perspective: a doubling of the squared Sharpe ratio is associated with a reduction in persistence of 0.86 to 0.33 percentage points, and the stability of this elasticity across specifications with and without firm controls suggests the relationship is not confounded by observable firm characteristics such as size, leverage, or sales growth.

The robustness of the result to the inclusion of industry fixed effects and firm controls in

Table 4: This table is the panel regression:

$$\text{Investment rate}_{i,t+1} = \mathbf{x}_{i,t}\boldsymbol{\xi} + \beta_1 \text{Volatility}_{i,t} + \beta_2 \text{InvDist}_t + \beta_3 \text{Volatility}_{i,t} \times \text{InvDist}_t + \delta_i + \gamma_{j,t} + \varepsilon_{i,t+1}$$

	(1)	(2)	(3)	(4)	(5)	(6)
Vol _{1y}	-0.70** (-2.34)	-0.76** (-2.51)	-5.27*** (-4.91)	-0.77*** (-2.77)	-0.82*** (-2.89)	-4.77*** (-3.91)
InvDist		-4.68*** (-5.37)	-7.39*** (-6.70)		-4.23*** (-4.61)	-6.62*** (-5.44)
Vol _{1y} × InvDist			5.57*** (4.21)			4.89*** (2.99)
Firm controls	Yes					
Fixed effects	Firm, Year		Firm, sic-2 × Year			
Standard errors	sic-2		sic-2, Year			
Adjusted R^2	0.51	0.51	0.51	0.52	0.52	0.53
Observations	81,714	81,714	81,714	81,543	81,543	81,543

t statistics in parentheses

* $p < 0.10$

** $p < 0.05$

*** $p < 0.01$

columns (3) and (4) carries particular economic significance. Industry fixed effects absorb systematic differences in investment persistence across sectors—capital-intensive industries exhibit slower adjustment for technological reasons unrelated to productivity dynamics, while service and technology firms adjust more rapidly—and the fact that the Sharpe ratio coefficient survives this industry-level control confirms that the relationship is not driven by the mechanical correlation between industry capital intensity, volatility, and persistence, but reflects genuine within-industry variation in the signal-to-noise ratio of individual firms' productivity processes. Similarly, the stability of the coefficient after including firm controls indicates that the Sharpe ratio is not simply proxying for investment opportunities or financial constraints—two alternative channels through which firms with strong productivity growth might exhibit faster mean reversion. Together, these results provide strong empirical support for the model's prediction that the spectral decay rate $\bar{\mu}^2/(2\sigma^2)$ is the sufficient statistic governing investment rate persistence.

To test the main prediction of this paper **H2**, I then run the following regression:

$$\begin{aligned} \text{Investment rate}_{i,t+1} = & \mathbf{x}_{i,t}\boldsymbol{\xi} + \beta_1 \text{Volatility}_{i,t} + \beta_2 \text{InvDist}_t \\ & + \beta_3 \text{Volatility}_{i,t} \times \text{InvDist}_t + \delta_i + \gamma_{j,t} + \varepsilon_{i,t+1} \end{aligned} \quad (35)$$

The coefficient of interest is β_3 in the above regression; it is the empirical test of Figure 1 and Corollary 1: high volatility is associated with higher average investment, but only if the firm is far away from the investment threshold.

The results in Table 4 establish three empirical regularities that are central to the paper’s argument. First, columns (1) and (4) confirm the baseline uncertainty-investment relationship: a one percentage-point increase in annual volatility is associated with a reduction in next-period investment rates of 0.70 to 0.82 percentage points, consistent with the real-options channel documented by the broader uncertainty literature. Second, and more importantly, columns (2) and (5) show that the model-implied investment distance measure—constructed as one minus the predicted probability of high investment from the out-of-sample probit—carries substantial incremental information about future investment rates beyond volatility alone. The coefficient of -4.68 to -4.23 implies that firms far from their investment threshold invest significantly less in the following period. Considering the within-firm standard deviation of corporate investment, a unit increase in distance (from a probability of 1 to a probability of 0) is associated with a 0.5 standard deviation decrease in firm-level investment. The robustness of this result to the inclusion of firm and industry-year fixed effects, which absorb all time-invariant firm heterogeneity and industry-wide demand shocks respectively, indicates that the distance measure captures genuine within-firm, within-industry variation in investment intensity rather than a spurious cross-sectional correlation.

The key result of the paper appears in columns (3) and (6), which introduce the interaction between volatility and investment distance. The positive and highly significant coefficient of 5.57 and 4.89 on $Vol_{1y} \times InvDist$ directly tests the sign-flip prediction of the model: the role of firm-level uncertainty on investment is heterogeneous across firms, governed by the distance to their investment thresholds. For firms close to their threshold - low $InvDist$, high predicted investment probability - the net effect of volatility is $\hat{\beta}_1 + \hat{\beta}_3 \times InvDist$, which at low values of $InvDist$ is dominated by the negative direct effect, consistent with the threshold displacement $\partial \hat{x} / \partial \sigma < 0$ and drift erosion $\partial \bar{\mu} / \partial \sigma < 0$ channels identified in the model’s decomposition. For firms far from their threshold, the positive interaction offsets the direct negative effect, capturing the diffusion benefit through which higher uncertainty opens paths to the threshold that drift alone cannot reach. The interaction coefficient of 4.89 in the most demanding specification - which includes firm fixed effects, industry-year fixed effects, and two-way clustered standard errors - survives the most stringent controls for unobserved heterogeneity, providing strong evidence that the sign reversal reflects the mechanism of the model.

While the previous table studied the cross-sectional relationship between the level of uncertainty and investment rates conditional on distance, Table 5 studies the casual effect of uncertainty shocks. In particular, I instrument the change in firm-level risk via two-stage least-squares (2SLS) using option-implied forward variance. In addition to being a forward-

Table 5: This table is the panel regression:

$$\text{Investment rate}_{i,t+1} = \mathbf{x}_{i,t}\boldsymbol{\xi} + \beta_1\Delta\text{Volatility}_{i,t} + \beta_2\text{InvDist}_t + \beta_3\Delta\text{Volatility}_{i,t} \times \text{InvDist}_t + \delta_i + \gamma_{j,t} + \varepsilon_{i,t+1}$$

The first three columns use the difference in firm-level historical volatility between $t - 1$ and t . In the next three columns, I instrument the change in volatility using rolling averages of 91-day simple model free implied variance based on [Martin 2017](#), provided by Grigory Vilkov's website. The use of 91 days is due to the lack of data at longer horizons.

	(1)	(2)	(3)	(4)	(5)	(6)
ΔVol_{1y}	-1.48*** (-4.90)	-1.51*** (-4.99)	-4.10*** (-3.18)			
InvDist		-3.64*** (-3.19)	-3.55*** (-3.07)		-5.46*** (-2.96)	-5.19*** (-2.71)
$\Delta\text{Vol}_{1y} \times \text{InvDist}$			3.20** (2.10)			
ΔVol_{1y} (2SLS)				-2.85*** (-2.71)	-3.01*** (-2.89)	-10.19*** (-3.43)
ΔVol_{1y} (2SLS) $\times$ InvDist						8.78*** (2.70)
Firm controls				Yes		
Fixed effects				Firm, Year		
Standard errors				sic-2		
Beginning Year		1984			1996	
Ending Year		2022			2019	
First-stage F -statistic		—		285.48	286.62	107.41
Adjusted R^2	0.51	0.51	0.51	0.17	0.17	0.17
Observations	48,426	48,426	48,426	15,873	15,873	15,873

t statistics in parentheses

* $p < 0.10$

** $p < 0.05$

*** $p < 0.01$

looking measure of uncertainty, this has the benefit of disentangling the collinearity between firm-level volatility and distance to investment, which are both estimated using historical data.

The first-differenced specifications in columns (1) through (3) speak to the dynamic investment response to changes in uncertainty rather than the static relationship between uncertainty levels and investment rates. Consistent with the previous table, we see that the same pattern holds: an increase in volatility is associated with a decrease in investment, but this relationship flips conditional on distance.

The instrumented specifications in columns (4) through (6) sharpen this economic interpretation by isolating the component of uncertainty that is exogenous to the firm’s environment. The amplification of the direct effect from -1.48 in the OLS to -2.85 to -3.01 in the 2SLS specifications is economically informative: it implies that realized volatility understates the true causal impact of uncertainty on investment by roughly half, consistent with the magnitudes found in [Alfaro et al. 2024](#). Importantly, the positive and statistically significant coefficient on the interaction term in column (6) indicates that the key mechanism presented in the paper survives empirically.

4 Conclusion

This paper presents a unifying framework to reconcile the conflicting facts in the investment and uncertainty literature, both in the time-series and in the cross-section. I derive closed-form solutions for the optimal investment rate and show using comparative statics how a simple statistic can be used to provide guidelines on when one can expect to see a positive or negative relationship.

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5 Appendix: Proofs

5.1 Proof of Lemma 2

Proof. By Dynkin's formula (see [Fleming and Soner 2006](#)),

$$\mathbb{E}_t[f(T, x)] = f(t, x) + \mathbb{E}_t \left[\int_t^T \left(\mathcal{D}(f(s, x)) + \frac{\partial f(s, x)}{\partial s} \right) \mathbf{d}s \right]$$

Using the boundary condition $f(T, x) = 0$ and the definition of $f(t, x)$,

$$\mathbb{E}_t \left[\int_t^T \left(\mathcal{D}(f^H(s, x)) + \frac{\partial f^H(s, x)}{\partial s} \right) \mathbf{d}s \right] = -\mathbb{E}_t \left[\int_t^T \mathbb{1}_{x_s \geq \hat{x}} \mathbf{d}s \right]$$

Bringing the expectations operator inside the integral via Fubini's theorem, applying the Radon-Nikodym theorem and by Lebesgue differentiation, we have

$$\mathcal{D}(f^H(t, x)) + \frac{\partial f^H(t, x)}{\partial t} = -\mathbb{1}_{x > \hat{x}} \quad \text{a.e.}$$

□

5.2 Proof of Proposition 1

Starting from

$$\frac{\partial f(t, x)}{\partial t} + \left[\mu - \frac{1}{2}\sigma^2 \right] \frac{\partial f(t, x)}{\partial x} + \frac{1}{2}\sigma^2 \frac{\partial^2 f(t, x)}{\partial x^2} = -\mathbb{1}_{x > \hat{x}}$$

and the terminal condition $f(T, x) = 0$, the goal is to find the solution to the Laplace transform

$$-\xi \bar{f}(\xi, x) + \left[\mu - \frac{1}{2}\sigma^2 \right] \frac{\partial \bar{f}(\xi, x)}{\partial x} + \frac{1}{2}\sigma^2 \frac{\partial^2 \bar{f}(\xi, x)}{\partial x^2} = -\frac{1}{\xi} \mathbb{1}_{x > \hat{x}}$$

with the change of variable $\tau = T - t$.

The ODE splits into two regions: $x < \hat{x}$ and $x > \hat{x}$. The general solution is given by

$$\bar{f}(\xi, x) = \begin{cases} A_1 e^{\lambda^+(\xi)x} + A_2 e^{\lambda^-(\xi)x} & \text{if } x < \hat{x} \\ \frac{1}{\xi^2} + B_1 e^{\lambda^+(\xi)x} + B_2 e^{\lambda^-(\xi)x} & \text{if } x > \hat{x} \end{cases}$$

where $\lambda^\pm(\xi) = -(\mu - 0.5\sigma^2)/\sigma^2 \pm \sqrt{(\mu - 0.5\sigma^2) + 2\sigma^2\xi}/\sigma^2$ and $\lambda^\pm(\xi)$ emphasizes the fact

that the solution to the characteristic equation is a function of the Laplace variable ξ . Economically, there are two boundary conditions,

$$\begin{aligned}\bar{f}(\xi, -\infty) &= 0 \\ \bar{f}(\xi, \infty) &= 1/\xi^2\end{aligned}$$

The first condition says that as x moves infinitely below $\hat{x}$, the expected occupation time above $\hat{x}$ is equal to zero. The second condition comes from the boundary condition $f(T - t, \infty) = T - t$ by a similar argument and applying the Laplace transform, $\int_0^\infty \tau e^{-\xi\tau} \mathbf{d}\tau = 1/\xi^2$. The two conditions imply

$$\bar{f}(\xi, x) = \begin{cases} A_1 e^{\lambda^+(\xi)x} & \text{if } x < \hat{x} \\ \frac{1}{\xi^2} + B_2 e^{\lambda^-(\xi)x} & \text{if } x > \hat{x} \end{cases}$$

and matching the value and derivative at $\hat{x}$,

$$\bar{f}(\xi, x) = \begin{cases} \frac{1}{2\xi^2} \left(1 + \frac{1}{\sqrt{1+2\left(\frac{\sigma}{\mu}\right)^2\xi}} \right) e^{\lambda^+(\xi)(x-\hat{x})} & \text{if } x < \hat{x} \\ \frac{1}{\xi^2} - \frac{1}{2\xi^2} \left(1 - \frac{1}{\sqrt{1+2\left(\frac{\sigma}{\mu}\right)^2\xi}} \right) e^{\lambda^-(\xi)(x-\hat{x})} & \text{if } x > \hat{x} \end{cases} \quad (36)$$

With explicit solutions for $\bar{f}(\xi, x)$, all that remains is to invert them. In general, the inverse Laplace transform is given by the Bromwich integral,

$$f(\tau, x) = \mathcal{L}^{-1}\{\bar{f}(\xi, x)\} = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{\xi\tau} \bar{f}(\xi, x) \mathbf{d}\xi \quad (37)$$

To proceed further, it is necessary to first state some results from complex analysis.

Remark 1. Let $f : D \rightarrow \mathbb{C}$ be an analytic function which has a pole at ξ_0 , and $D \subset \mathbb{C}$. Then,

$$\lim_{\xi \rightarrow \xi_0} |f(\xi)| = \infty.$$

For our purposes, we can consider the definition of a pole as a point at which the function is not well-behaved, and singularities in general as such points. Moreover,

Definition 1 (Residue). Let $f : \dot{U}_r(\xi_0) \rightarrow \mathbb{C}$ be an analytic function, such that ξ_0 is a

singularity of f , and let

$$f(\xi) = \sum_{n=-\infty}^{\infty} a_n(\xi - \xi_0)^n$$

be its Laurent series in $\dot{U}_r(\xi_0)$ where $\dot{U}_r(\xi_0) \doteq \{\xi \in \mathbb{C} : 0 < |\xi - \xi_0| < r\}$ is a punctured disk centered at ξ_0 . The coefficient a_{-1} in this expansion is called the residue of f at the point ξ_0 .

For well-behaved functions that do not possess singularities, the coefficients $a_{-1}, a_{-2}, \dots$ simply equal zero and the above representation reduces to its Taylor series. Finally,

Theorem 4 (The Residue Theorem). Let $f(\xi)$ be analytic inside and on a simple closed contour Ω , except for a finite number of isolated singular points $\xi_1, \xi_2, \dots, \xi_N$ located inside Ω . Then,

$$\oint_{\Omega} f(\xi) d\xi = 2\pi i \sum_j^N \text{Res}\{f(\xi); \xi_j\}$$

Together, the above definitions and results tell us that if we can successfully close the path of integration for the Bromwich contour in Equation (37), we can further analyze its properties by simply evaluating the behavior of the integrand at its singular points.¹⁰ Proposition 1 then applies the residue theorem (among other things) to our setting.

By inspection of Equation (36), $\bar{f}(\xi, x)$ has a (second-order) pole at $\xi_0 = 0$. As a result, we can choose an arbitrary value of $\gamma > 0$ and close the contour along the left-axis of the complex plane while enclosing the point ξ_0 . Figure IA.8 illustrates the path of integration.

Importantly, we need to ensure the analyticity of $\bar{f}(\xi, x)$ in and on the closed contour Ω . Because $\bar{f}(\xi, x)$ involves the square root of a complex number ξ in the characteristic equation for $\lambda(\xi)$, it is in fact multi-valued for values of $\xi \neq -\bar{\mu}^2/(2\sigma^2)$ as the function approaches the real axis. This will hold more broadly whenever the original function $f(\cdot)$ solves a linear second-order parabolic equation as is often the case in Finance. To address this issue, given the placement of the Bromwich line γ , the location of the pole ξ_0 and the branch point ξ_b , we can simply cut the plane along the negative real axis and loop around ξ_b as shown in Figure IA.8.

¹⁰Evaluation of the Bromwich integral via the Residue theorem does not require the function $\bar{f}(\xi, x)$ to possess poles. If the function is analytic everywhere or can be analytically extended by continuation onto $D \cup \{\xi_0\}$, Cauchy's theorem applies and the sum of residues equal zero. In this case, branch points, which will generally exist in our setting, will suffice to evaluate the Bromwich integral meaningfully.

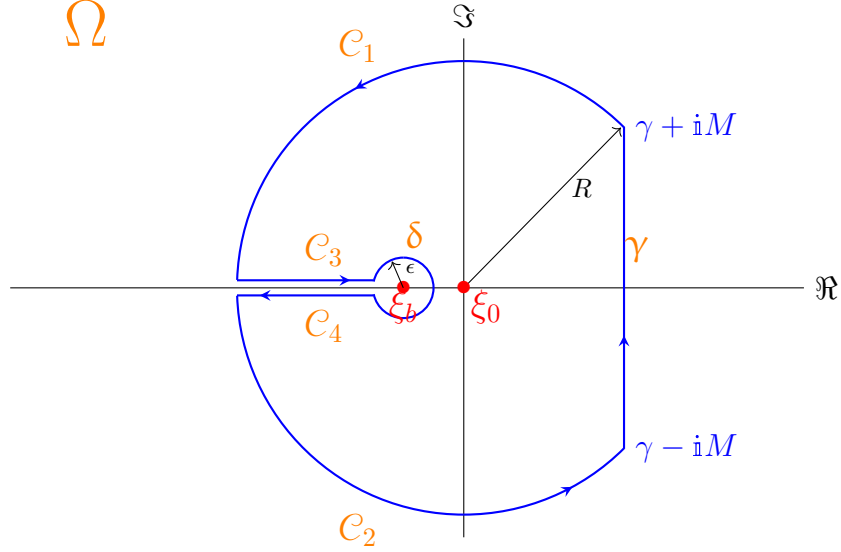


Figure IA.8: **Path of integration with a branch cut around ξ_b and a pole at ξ_0 .** This figure plots the closed contour Ω which connects 6 separate contours: $C_1, C_2, C_3, C_4, \delta, \gamma$. The points ξ_0, ξ_b indicate the second-order pole and branch point of the function $\bar{f}(\xi, x)$, and ϵ, R indicate the radius of the semi-circular arcs enclosing these points.

By the Residue theorem,

$$\oint_{\Omega} e^{\xi\tau} \bar{f}(\xi, x) d\xi = 2\pi i \times \text{Res}\{e^{\xi\tau} \bar{f}(\xi, x); \xi_0\}$$

but also,

$$\oint_{\Omega} = \int_{C_1} + \int_{C_2} + \int_{C_3} + \int_{C_4} + \int_{\delta} + \int_{\gamma}$$

Clearly, $\int_{\gamma}$ is our object of interest. To match the Bromwich integral in Equation (37), we simply take limits as $R \rightarrow \infty$, such that our vertical line γ in Figure IA.8 extends from $\gamma - i\infty$ to $\gamma + i\infty$.

In the other direction, we can take $\epsilon \rightarrow 0$ such that C_3 and C_4 approach the real-axis from above and below.

Then, by Jordan's lemma,

$$\begin{aligned} \oint_{\Omega} &= \lim_{\substack{R \rightarrow \infty \\ \epsilon \rightarrow 0}} \int_{C_1} + \int_{C_2} + \int_{C_3} + \int_{C_4} + \int_{\delta} + \int_{\gamma} \\ &= \int_{C_3} + \int_{C_4} + \int_{\gamma} = \lim_{\epsilon \rightarrow 0^+} \int_{-\infty}^{\xi_b} + \lim_{\epsilon \rightarrow 0^-} \int_{\xi_b}^{-\infty} + \int_{\gamma - i\infty}^{\gamma + i\infty} \end{aligned}$$

which implies that

$$\int_{\gamma-i\infty}^{\gamma+i\infty} = \oint_{\Omega} - \lim_{\epsilon \rightarrow 0^+} \int_{C_3} - \lim_{\epsilon \rightarrow 0^-} \int_{C_4}$$

or,

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{\xi\tau} \bar{f}(\xi, x) d\xi &= \text{Res}\{e^{\xi\tau} \bar{f}(\xi, x); \xi_0\} \\ &- \frac{1}{2\pi i} \left(\lim_{\epsilon \rightarrow 0^+} \int_{-\infty}^{\xi_b} e^{\xi\tau} \bar{f}(\xi, x) d\xi + \lim_{\epsilon \rightarrow 0^-} \int_{\xi_b}^{-\infty} e^{\xi\tau} \bar{f}(\xi, x) d\xi \right) \end{aligned}$$

Case: $x < \hat{x}$:

To proceed, I first compute the residue at $\xi_0 = 0$, then calculate the branch cut contribution separately. The term $e^{\xi\tau} \bar{f}(\xi, x)$ can be written,

$$e^{\xi\tau} \bar{f}(\xi, x) = \frac{1}{2\xi^2} \left(1 + \frac{1}{\sqrt{1 + 2\left(\frac{\sigma}{\bar{\mu}}\right)^2 \xi}} \right) \times \exp \left(\xi\tau + \left(\frac{-\bar{\mu}}{\sigma^2} + \frac{2\sqrt{\bar{\mu} + 2\sigma^2\xi}}{\sigma^2} \right) (x - \hat{x}) \right)$$

To compute the residue at ξ_0 , we express $e^{\xi\tau} \bar{f}(\xi, x)$ in its Laurent expansion around ξ_0 and find the coefficient of $\frac{1}{\xi - \xi_0}$, corresponding to the a_{-1} term in the definition of a residue. Expanding term by term,

$$\exp \left[\left(\frac{-\bar{\mu}}{\sigma^2} + \frac{2\sqrt{\bar{\mu} + 2\sigma^2\xi}}{\sigma^2} \right) (x - \hat{x}) \right] \approx 1 - \frac{2(\hat{x} - x)}{\bar{\mu}} \xi + \frac{2(\hat{x} - x)(\sigma^2\bar{\mu} + 2\bar{\mu}^2(\hat{x} - x))}{\bar{\mu}^4} \xi^2 + \dots$$

$$\frac{1}{2} \left(1 + \frac{1}{\sqrt{1 + 2\left(\frac{\sigma}{\bar{\mu}}\right)^2 \xi}} \right) \approx 1 - \frac{1}{2} \left(\frac{\sigma}{\bar{\mu}} \right)^2 \xi + \frac{3}{4} \left(\frac{\sigma}{\bar{\mu}} \right)^4 \xi^2 - \frac{5}{4} \left(\frac{\sigma}{\bar{\mu}} \right)^6 \xi^3 + \dots$$

$$\exp(\xi\tau) \approx 1 + \tau\xi + \frac{1}{2}\tau^2\xi^2 + \frac{1}{6}\tau^3\xi^3 + \dots$$

Collecting and simplifying, we have

$$\text{Res}\{e^{\xi\tau} \bar{f}(\xi, x); \xi_0\} = \tau - \frac{(\hat{x} - x)}{\bar{\mu}} - \frac{1}{2} \frac{\sigma^2}{\bar{\mu}^2}$$

Next, I consider the integral: $\lim_{\epsilon \rightarrow 0^+} \frac{1}{2\pi\mathbf{i}} \int_{-\infty}^{\xi_b} e^{\xi\tau} \bar{f}(\xi, x) \mathbf{d}\xi$,

$$= \lim_{\epsilon \rightarrow 0^+} \frac{1}{2\pi\mathbf{i}} \int_{-\infty}^{\xi_b} \frac{1}{2\xi^2} \left(1 + \frac{1}{\sqrt{1 + 2\left(\frac{\sigma}{\bar{\mu}}\right)^2} \xi} \right) \times \exp \left(\xi\tau + \left(\frac{-\bar{\mu}}{\sigma^2} + \frac{2\sqrt{\bar{\mu} + 2\sigma^2\xi}}{\sigma^2} \right) (x - \hat{x}) \right) \mathbf{d}\xi$$

Since $\xi_b = -\bar{\mu}/(2\sigma^2)$ and the branch cut runs along $(-\infty, \xi_b)$, we must express ξ in polar coordinates inside square roots. Taking $\arg(z) \in (-\pi, \pi)$,

$$\begin{aligned} &= \frac{1}{2\pi\mathbf{i}} \int_{-\infty}^{\xi_b} \frac{1}{2\xi^2} \left(1 + \frac{1}{\sqrt{1 + 2\left(\frac{\sigma}{\bar{\mu}}\right)^2} |\xi| e^{\mathbf{i}\pi}} \right) \times \exp \left(\xi\tau + \left(\frac{-\bar{\mu}}{\sigma^2} + \frac{2\sqrt{\bar{\mu} + 2\sigma^2|\xi|} e^{\mathbf{i}\pi}}{\sigma^2} \right) (x - \hat{x}) \right) \mathbf{d}\xi \\ &= \frac{1}{2\pi\mathbf{i}} \int_{-\infty}^0 \frac{1}{2(\xi + \xi_b)^2} \left(1 - \frac{\mathbf{i}}{\frac{\sigma}{\bar{\mu}} \sqrt{2|\xi|}} \right) \times \exp \left((\xi + \xi_b)\tau + \left(\frac{-\bar{\mu}}{\sigma^2} + \frac{2\mathbf{i}\sqrt{2|\xi|}}{\sigma} \right) (x - \hat{x}) \right) \mathbf{d}\xi \\ &= \frac{1}{2\pi\mathbf{i}} \int_{-\infty}^0 \frac{1}{2(\xi_b - \xi)^2} \left(1 - \frac{\mathbf{i}}{\frac{\sigma}{\bar{\mu}} \sqrt{2\xi}} \right) \times \exp \left((\xi_b - \xi)\tau + \left(\frac{-\bar{\mu}}{\sigma^2} + \frac{2\mathbf{i}\sqrt{2\xi}}{\sigma} \right) (x - \hat{x}) \right) \mathbf{d}\xi \\ &= -\frac{1}{2\pi\mathbf{i}} \int_0^{\infty} \frac{1}{2(\xi_b - \xi)^2} \left(1 - \frac{\mathbf{i}}{\frac{\sigma}{\bar{\mu}} \sqrt{2\xi}} \right) \times \exp \left((\xi_b - \xi)\tau + \left(\frac{-\bar{\mu}}{\sigma^2} + \frac{2\mathbf{i}\sqrt{2\xi}}{\sigma} \right) (x - \hat{x}) \right) \mathbf{d}\xi \end{aligned}$$

Doing the same for the integral: $\lim_{\epsilon \rightarrow 0^-} \frac{1}{2\pi\mathbf{i}} \int_{\xi_b}^{-\infty} e^{\xi\tau} \bar{f}(\xi, x) \mathbf{d}\xi$,

$$= \frac{1}{2\pi\mathbf{i}} \int_0^{\infty} \frac{1}{2(\xi_b - \xi)^2} \left(1 + \frac{\mathbf{i}}{\frac{\sigma}{\bar{\mu}} \sqrt{2\xi}} \right) \times \exp \left((\xi_b - \xi)\tau + \left(\frac{-\bar{\mu}}{\sigma^2} - \frac{2\mathbf{i}\sqrt{2\xi}}{\sigma} \right) (x - \hat{x}) \right) \mathbf{d}\xi$$

Taking the difference and using $(1 - a) \exp(\mathbf{i}b) - (1 + a) \exp(-\mathbf{i}b) = 2\mathbf{i} \sin(b) - 2a \cos(b)$,

$$\begin{aligned} &- \frac{1}{2\pi\mathbf{i}} \left(\lim_{\epsilon \rightarrow 0^+} \int_{-\infty}^{\xi_b} e^{\xi\tau} \bar{f}(\xi, x) \mathbf{d}\xi + \lim_{\epsilon \rightarrow 0^-} \int_{\xi_b}^{-\infty} e^{\xi\tau} \bar{f}(\xi, x) \mathbf{d}\xi \right) \\ &= \frac{1}{2\pi} \int_0^{\infty} \frac{1}{\left(\frac{\bar{\mu}^2}{2\sigma^2} + \xi\right)^2} e^{-\left(\frac{\bar{\mu}^2}{2\sigma^2} + \xi\right)\tau - \frac{\bar{\mu}}{\sigma^2}(x - \hat{x})} \times \left[\sin \left(\frac{\sqrt{2\xi}}{\sigma} (\hat{x} - x) \right) + \frac{\bar{\mu}}{\sigma\sqrt{2\xi}} \cos \left(\frac{\sqrt{2\xi}}{\sigma} (\hat{x} - x) \right) \right] \mathbf{d}\xi \end{aligned}$$

Adding the residue and the branch cut contribution concludes the derivation of the case when $x < \hat{x}$. Applying the same steps for the solution to the ODE in Equation (36) for the case $x > \hat{x}$ concludes the proof.

5.3 Proof of Proposition 2

$$\begin{aligned}
rV^H(z, k, A) &= \zeta_0 A + (\mu - \kappa \sigma^H) A \frac{\partial V(z, k, A)}{\partial A} + \frac{1}{2} (\sigma^H)^2 A^2 \frac{\partial^2 V(z, k, A)}{\partial A^2} \\
&\quad + \left[V^H(z + \tilde{\mathcal{J}}, k + \tilde{\mathcal{K}}, A) - V^H(z, k, A) - q \right] + \eta^L [V^L(z, k, A) - V^H(z, k, A)] \\
rV^L(z, k, A) &= \zeta_0 A + (\mu - \kappa \sigma^L) A \frac{\partial V(z, k, A)}{\partial A} + \frac{1}{2} (\sigma^L)^2 A^2 \frac{\partial^2 V(z, k, A)}{\partial A^2} \\
&\quad + \left[V^L(z + \tilde{\mathcal{J}}, k + \tilde{\mathcal{K}}, A) - V^L(z, k, A) - q \right] + \eta^H [V^H(z, k, A) - V^L(z, k, A)]
\end{aligned}$$

with boundary conditions

$$\begin{aligned}
V^H(z, k, A^{*H}) &= V^H(z + \tilde{\mathcal{J}}, k + \tilde{\mathcal{K}}, A^{*H}) - q \\
V^L(z, k, A^{*H}) &= V^L(z + \tilde{\mathcal{J}}, k + \tilde{\mathcal{K}}, A^{*H}) - q
\end{aligned}$$

Applying the change of variable $x = \log A$ and defining the following:

$$\nu^H \doteq V_x^H \quad ; \quad \nu^L \doteq V_x^L$$

The ODE can be written as:

$$\begin{aligned}
\nabla \begin{bmatrix} V^H \\ \nu^H \\ V^L \\ \nu^L \end{bmatrix} &= \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{(1+r+\eta^L)}{.5(\sigma^H)^2} & -\frac{\mu-\kappa\sigma^H-.5(\sigma^H)^2}{.5(\sigma^H)^2} & -\frac{\eta^L}{.5(\sigma^H)^2} & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{\eta^H}{.5(\sigma^L)^2} & 0 & \frac{(1+r+\eta^H)}{.5(\sigma^L)^2} & -\frac{\mu-\kappa\sigma^L-.5(\sigma^L)^2}{.5(\sigma^L)^2} \end{bmatrix}}_{\doteq M} \begin{bmatrix} V^H \\ \nu^H \\ V^L \\ \nu^L \end{bmatrix} + \begin{bmatrix} 0 \\ \left[\zeta_0 + \frac{\tilde{\zeta}_1}{\sigma^H} \right] e^s - q \\ 0 \\ \left[\zeta_0 + \frac{\tilde{\zeta}_1}{\sigma^L} \right] e^s - q \end{bmatrix}
\end{aligned} \tag{38}$$

The general solution to system of constant coefficient first order linear equations in Equation (8) is

$$\mathbf{V} = \sum_i^4 c_i \mathbf{v}^{(i)} e^{\omega^{(i)} x}$$

where c_i are integration constants and $\mathbf{v}^{(i)}$ is the eigenvector associated with the i th eigenvalue $\omega^{(i)}$.

In principle, the eigenvalues can be computed explicitly from $\det(M - \omega I) = 0$ using Ferrari's method. Though the main body of the text uses exact solutions, for analytic convenience, we can treat M in a block structure which leads to an approximate error of

$(\frac{\eta^H}{.5(\sigma^L)^2})(\frac{\eta^L}{.5(\sigma^H)^2})$. Then, the eigenvalues satisfy

$$\left(\omega^2 + \frac{\mu - \kappa\sigma^H - .5(\sigma^H)^2}{.5(\sigma^H)^2}\omega - \frac{(r+1+\eta^L)}{.5(\sigma^H)^2}\right) \left(\omega^2 + \frac{\mu - \kappa\sigma^L - .5(\sigma^L)^2}{.5(\sigma^L)^2}\omega - \frac{(r+1+\eta^H)}{.5(\sigma^L)^2}\right) = 0$$

We have 4 solutions for ω . However, we require $e^{\omega x} \rightarrow 0$ as $x \rightarrow -\infty$ so we are left with two positive solutions,

$$\omega^{(1)} = -\frac{\mu - \kappa\sigma^H - .5(\sigma^H)^2}{(\sigma^H)^2} + \sqrt{\left(\frac{\mu - \kappa\sigma^H - .5(\sigma^H)^2}{.5(\sigma^H)^2}\right)^2 + 4\left(\frac{r+1+\eta^L}{.5(\sigma^H)^2}\right)} / 2$$

and

$$\omega^{(3)} = -\frac{\mu - \kappa\sigma^L - .5(\sigma^L)^2}{(\sigma^L)^2} + \sqrt{\left(\frac{\mu - \kappa\sigma^L - .5(\sigma^L)^2}{.5(\sigma^L)^2}\right)^2 + 4\left(\frac{r+1+\eta^H}{.5(\sigma^L)^2}\right)} / 2$$

The eigenvector associated with the i th eigenvalue is

$$\mathbf{v}^{(i)} = \begin{bmatrix} v_1^{(i)} \\ v_2^{(i)} \\ v_3^{(i)} \\ v_4^{(i)} \end{bmatrix} = \begin{bmatrix} v_1^{(i)} \\ \omega^{(i)} v_1^{(i)} \\ v_1^{(i)} \frac{r+1+\eta^L - (\mu - \kappa\sigma^H - .5(\sigma^H)^2 + \omega^{(i)} / (.5(\sigma^H)^2)) \omega^{(i)}}{\eta^L} \\ \omega^{(i)} v_3^{(i)} \end{bmatrix}$$

Collecting, the value functions V^H, V^L are given by

$$\begin{aligned} V^H &= c_1 v_1^{(1)} e^{\omega^{(1)} x} + c_3 v_1^{(3)} e^{\omega^{(3)} x} + a_1 e^x - \frac{1}{r+1} q \\ V^L &= c_1 v_3^{(1)} e^{\omega^{(1)} x} + c_3 v_3^{(3)} e^{\omega^{(3)} x} + a_3 e^x - \frac{1}{r+1} q \end{aligned}$$

where

$$\begin{aligned} a_1 &= \frac{\eta^L \left(\zeta_0 + \frac{\tilde{\zeta}_1}{\delta^L} \right) + \Gamma^L \left(\zeta_0 + \frac{\tilde{\zeta}_1}{\delta^H} \right)}{\Gamma^L \Gamma^H - \eta^H \eta^L} \\ a_3 &= \frac{\eta^H \left(\zeta_0 + \frac{\tilde{\zeta}_1}{\delta^H} \right) + \Gamma^H \left(\zeta_0 + \frac{\tilde{\zeta}_1}{\delta^L} \right)}{\Gamma^L \Gamma^H - \eta^H \eta^L} \end{aligned}$$

and I define $\Gamma^H \doteq 1 + r + \eta^L - \mu + \kappa\sigma^H$ and $\Gamma^L \doteq 1 + r + \eta^H - \mu + \kappa\sigma^L$. Moreover, let $\zeta_3 \doteq [\alpha(z + \tilde{z} + \tilde{\tilde{z}})^\varepsilon + (1 - \alpha)(k + \tilde{k} + \tilde{\tilde{k}})^\varepsilon]^{1/\varepsilon}$. Applying the boundary conditions, the integration

constants c_1 and c_3 are pinned down by

$$c_1 = \frac{e^{-\omega^{(1)}\hat{x}} \left[v_1^{(3)} ((b_3 - a_3)e^{\hat{x}} - q) - v_3^{(3)} ((b_1 - a_1)e^{\hat{x}} - q) \right]}{v_1^{(3)}v_3^{(1)} - v_1^{(1)}v_3^{(3)}}$$

$$c_3 = \frac{e^{-\omega^{(3)}\check{x}} \left[v_3^{(1)} ((b_1 - a_1)e^{\check{x}} - q) - v_1^{(1)} ((b_3 - a_3)e^{\check{x}} - q) \right]}{v_1^{(3)}v_3^{(1)} - v_1^{(1)}v_3^{(3)}}$$

where

$$b_1 = \frac{\eta^L \left(\zeta_1 + \frac{\tilde{\zeta}_2}{\delta^L} \right) + \Gamma^L \left(\zeta_1 + \frac{\tilde{\zeta}_2}{\delta^H} \right)}{\Gamma^L \Gamma^H - \eta^H \eta^L}$$

$$b_3 = \frac{\eta^H \left(\zeta_1 + \frac{\tilde{\zeta}_2}{\delta^H} \right) + \Gamma^H \left(\zeta_1 + \frac{\tilde{\zeta}_2}{\delta^L} \right)}{\Gamma^L \Gamma^H - \eta^H \eta^L}$$

and $\hat{x}, \check{x}$ denote the optimal investment thresholds in the H and L regimes respectively.

5.4 Proof of Lemma 3

Proof. Due to symmetry, I focus on $f^H(t, x)$. By Dynkin's formula,

$$\mathbb{E}_t [f^H(T, x)] = f^H(t, x) + \mathbb{E}_t \left[\int_t^T \left(\mathcal{D}(f^H(s, x)) + \frac{\partial f^H(s, x)}{\partial s} + \eta^L [f^L(t, x) - f^H(t, x)] \right) \mathbf{d}s \right]$$

using the boundary condition $f^H(T, x) = 0$ and the definition of $f^H(t, x)$,

$$\mathbb{E}_t \left[\int_t^T \left(\mathcal{D}(f^H(s, x)) + \frac{\partial f^H(s, x)}{\partial s} + \eta^L [f^L(t, x) - f^H(t, x)] \right) \mathbf{d}s \right] = -\mathbb{E}_t \left[\int_0^T \mathbb{1}_{x_s \geq \hat{x}^H} \mathbf{d}s \right]$$

Bringing the expectations operator inside the integral via Fubini's theorem, applying the Radon-Nikodym theorem and by Lebesgue differentiation, we have

$$\mathcal{D}(f^H(t, x)) + \frac{\partial f^H(t, x)}{\partial t} + \eta^L [f^L(t, x) - f^H(t, x)] = -\mathbb{1}_{x > \hat{x}^H} \quad \text{a.e.}$$

□

5.5 Proof of Proposition 3

Starting with the system of PDEs:

$$\begin{aligned}\frac{\partial f^H(t, x)}{\partial t} + \left[\mu^H - \frac{1}{2}(\sigma^H)^2 \right] \frac{\partial f^H(t, x)}{\partial x} + \frac{1}{2}(\sigma^H)^2 \frac{\partial^2 f^H(t, x)}{\partial x^2} + \eta^L \left[f^L(t, x) - f^H(t, x) \right] &= -\mathbb{1}_{x > \hat{x}^H} \\ \frac{\partial f^L(t, x)}{\partial t} + \left[\mu^L - \frac{1}{2}(\sigma^L)^2 \right] \frac{\partial f^L(t, x)}{\partial x} + \frac{1}{2}(\sigma^L)^2 \frac{\partial^2 f^L(t, x)}{\partial x^2} + \eta^H \left[f^H(t, x) - f^L(t, x) \right] &= -\mathbb{1}_{x > \hat{x}^L}\end{aligned}$$

I reduce the above system into a system of ODEs via the Laplace transform, $\bar{f}^H(\xi, x) = \mathcal{L}\{f^H\} \doteq \int_0^\infty e^{-\xi t} f^H(t, x) \mathbf{d}t$. The system can be rewritten:

$$\begin{aligned}\xi \bar{f}^H(\xi, x) + \left[\mu^H - \frac{1}{2}(\sigma^H)^2 \right] \frac{\partial \bar{f}^H(\xi, x)}{\partial x} + \frac{1}{2}(\sigma^H)^2 \frac{\partial^2 \bar{f}^H(\xi, x)}{\partial x^2} + \eta^L \left[\bar{f}^L(\xi, x) - \bar{f}^H(\xi, x) \right] &= -\frac{1}{\xi} \mathbb{1}_{x > \hat{x}^H} \\ \xi \bar{f}^L(\xi, x) + \left[\mu^L - \frac{1}{2}(\sigma^L)^2 \right] \frac{\partial \bar{f}^L(\xi, x)}{\partial x} + \frac{1}{2}(\sigma^L)^2 \frac{\partial^2 \bar{f}^L(\xi, x)}{\partial x^2} + \eta^H \left[\bar{f}^H(\xi, x) - \bar{f}^L(\xi, x) \right] &= -\frac{1}{\xi} \mathbb{1}_{x > \hat{x}^L}\end{aligned}$$

As we have seen, whether $\hat{x}^H > \hat{x}^L$ or $\hat{x}^H < \hat{x}^L$ can depend on the values of μ^H and μ^L . I focus on more common case in which $\hat{x}^H > \hat{x}^L$, i.e., the no-action region expands in the high volatility regime. Then, the system of ODEs is split into three regions: $x < \hat{x}^L$, $\hat{x}^L < x < \hat{x}^H$ and $x > \hat{x}^H$.

As before, the general solution is given by:

$$\mathbf{F} = \sum_i^4 c_i \mathbf{v}^{(i)} e^{\lambda^{(i)} x}$$

where c_i are integration constants and $\mathbf{v}^{(i)}$ is the eigenvector associated with the i th eigenvalue $\lambda^{(i)}$. The eigenvalues are the solutions to the quartic equation

$$\left(\lambda^2 + \frac{\mu^H - .5(\sigma^H)^2}{.5(\sigma^H)^2} \lambda + \frac{\xi - \eta^L}{.5(\sigma^H)^2} \right) \left(\lambda^2 + \frac{\mu^L - .5(\sigma^L)^2}{.5(\sigma^L)^2} \lambda - \frac{\xi - \eta^H}{.5(\sigma^L)^2} \right) - \left(\frac{\eta^H}{.5(\sigma^H)^2} \right) \left(\frac{\eta^L}{.5(\sigma^L)^2} \right) = 0$$

which can be solved explicitly. Solving the ODEs piecewise and matching at $x = \hat{x}^H$ and at $x = \hat{x}^L$, we have the following solution:

$x < \hat{x}^L$:

$$\bar{f}^H(\xi, x) = c_1^H e^{\lambda^{(1)}(x-\hat{x}^L)} + c_3^H e^{\lambda^{(3)}(x-\hat{x}^L)}$$

$$\bar{f}^L(\xi, x) = c_1^L e^{\lambda^{(1)}(x-\hat{x}^L)} + c_3^L e^{\lambda^{(3)}(x-\hat{x}^L)}$$

$\hat{x}^L < x < \hat{x}^H$:

$$\bar{f}^H(\xi, x) = \mathfrak{c}_1^H e^{\lambda^{(1)}(x-\hat{x}^H)} + \mathfrak{c}_2^H e^{\lambda^{(2)}(x-\hat{x}^L)} + \mathfrak{c}_3^H e^{\lambda^{(3)}(x-\hat{x}^H)} + \mathfrak{c}_4^H e^{\lambda^{(4)}(x-\hat{x}^L)} + \frac{\eta^L}{\xi^2(\xi - \eta^H - \eta^L)}$$

$$\bar{f}^L(\xi, x) = \mathfrak{c}_1^L e^{\lambda^{(1)}(x-\hat{x}^H)} + \mathfrak{c}_2^L e^{\lambda^{(2)}(x-\hat{x}^L)} + \mathfrak{c}_3^L e^{\lambda^{(3)}(x-\hat{x}^H)} + \mathfrak{c}_4^L e^{\lambda^{(4)}(x-\hat{x}^L)} - \frac{\xi - \eta^L}{\xi^2(\xi - \eta^H - \eta^L)}$$

$\hat{x}^H < x$:

$$\bar{f}^H(\xi, x) = \mathfrak{c}_2^H e^{\lambda^{(2)}(x-\hat{x}^H)} + \mathfrak{c}_4^H e^{\lambda^{(4)}(x-\hat{x}^H)} - \frac{1}{\xi^2}$$

$$\bar{f}^L(\xi, x) = \mathfrak{c}_2^L e^{\lambda^{(2)}(x-\hat{x}^H)} + \mathfrak{c}_4^L e^{\lambda^{(4)}(x-\hat{x}^H)} - \frac{1}{\xi^2}$$

The 16 integration constants have explicit forms and can be found by imposing matching at $x = \hat{x}^H$ and at $x = \hat{x}^L$, along with boundary conditions at $x \rightarrow -\infty$ and $x \rightarrow \infty$ (for example, $\bar{f}^L(\xi, -\infty) = 0$). Moreover, $\lambda^{(1)}, \lambda^{(3)}$ are the positive eigenvalues to the characteristic equation while $\lambda^{(2)}, \lambda^{(4)}$ are the negative ones.

With explicit solutions for $\bar{f}^H(\xi, x)$ and $\bar{f}^L(\xi, x)$, all that remains is to invert them. Remarkably, the poles lie at $\xi = 0$ and $\xi = \eta^H + \eta^L$. Hence, the expected occupation time can be found by evaluating the behavior of $\bar{f}$ at these two points, which can easily be done numerically.

6 Appendix: Data

6.1 Bootstrap Procedure

To account for uncertainty in the first-stage estimates of investment persistence, I employ a parametric residual bootstrap. In the first stage, I estimate firm-level AR(1) regressions once on the original data, yielding the estimates $\hat{\rho}_i$, $\hat{\alpha}_i$, and centred residuals $\hat{\varepsilon}_{i,t} = \text{IR}_{i,t} - \hat{\alpha}_i - \hat{\rho}_i \text{IR}_{i,t-1}$ where IR stands for Investment Rate. In each of $B = 500$ replications, I then draw N firms with replacement. For each resampled firm, I construct a synthetic investment rate series by drawing residuals with replacement from that firm's own empirical residual distribution and recursively rebuild the series as $\widetilde{\text{IR}}_{i,t} = \hat{\alpha}_i + \hat{\rho}_i \widetilde{\text{IR}}_{i,t-1} + \varepsilon_{i,t}^*$, where $\varepsilon_{i,t}^* \stackrel{\text{iid}}{\sim} \hat{F}_i$ and $\hat{F}_i$ is the empirical distribution of firm i 's residuals. I then re-estimate persistence on the synthetic series, yielding $\hat{\rho}_i^{(b)}$. Next, I recompute the within-year Sharpe ratio ranks on the resampled cross-section. The second-stage cross-sectional regression is run at each t using $\hat{\rho}_i^{(b)}$ as the dependent variable, and coefficients are averaged across years to yield $\hat{\beta}^{(b)}$. The bootstrap standard error is the standard deviation of $\{\hat{\beta}^{(b)}\}_{b=1}^B$.

The procedure is formalized as follows. The first stage is estimated once on the original data as before, with centered residuals

$$\hat{\varepsilon}_{i,t} \leftarrow \text{IR}_{i,t} - \hat{\alpha}_i - \hat{\rho}_i \text{IR}_{i,t-1}. \quad (39)$$

For each replication $b = 1, \dots, B$:

Step 1: $\mathcal{S}^{(b)} = \text{draw } N \text{ firms with replacement from } \{1, \dots, N\},$

Step 2: $\varepsilon_{i,t}^* \stackrel{\text{iid}}{\sim} \hat{F}_i \quad \forall i \in \mathcal{S}^{(b)},$

Step 3: $\widetilde{\text{IR}}_{i,t} = \hat{\alpha}_i + \hat{\rho}_i \widetilde{\text{IR}}_{i,t-1} + \varepsilon_{i,t}^*,$

Step 4: $\hat{\rho}_i^{(b)} = \text{AR}(1)(\widetilde{\text{IR}}_{i,\cdot}),$

Step 5: $\widetilde{\text{SR}}_{i,t}^{(b)} = \text{Rank}(\text{SR}_{i,t}^2 \mid i \in \mathcal{S}^{(b)}, t) \quad \forall t,$

Step 6: $\hat{\beta}_t^{(b)} = \text{CS-reg}(\hat{\rho}_i^{(b)}, \widetilde{\text{SR}}_{i,t}^{(b)}, \mathbf{x}_{i,t} \mid i \in \mathcal{S}^{(b)}) \quad \forall t,$

Step 7: $\hat{\beta}^{(b)} = \frac{1}{T} \sum_{t=1}^T \hat{\beta}_t^{(b)}.$

The bootstrap standard error and t -statistic are as in the main table. Two additional design

choices are worth noting. First, residuals are resampled nonparametrically: no distributional assumption is imposed on $\hat{F}_i$, so fat tails and skewness in firm-level investment shocks are preserved. Second, the original estimate $\hat{\rho}_i$ is used to generate the synthetic series in Step 3 but re-estimated in Step 4. This ensures the data-generating process is centered at the original estimate while the re-estimation in Step 4 recovers the distribution of $\hat{\rho}_i^{(b)}$ around $\hat{\rho}_i$, which addresses the fact that $\hat{\rho}_i$ is an estimated dependent variable.

7 Appendix: Supplementary Figures

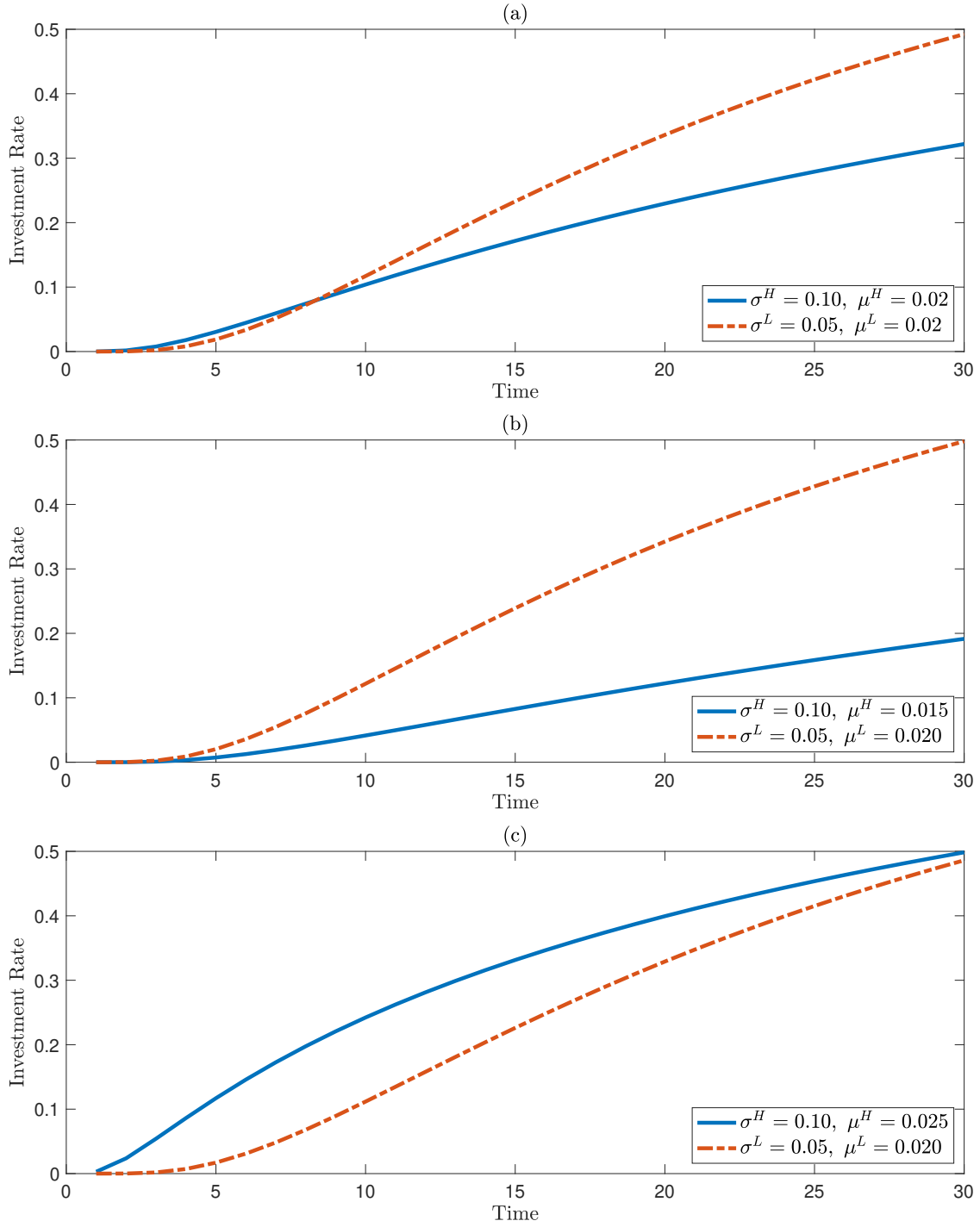


Figure IA.9: Investment rate with stochastic volatility as a function of the time span, fixing the initial point at $x_0 = 0$. This figure studies the following cases: $\mu^H = \mu^L$, $\mu^H > \mu^L$, and $\mu^H < \mu^L$.

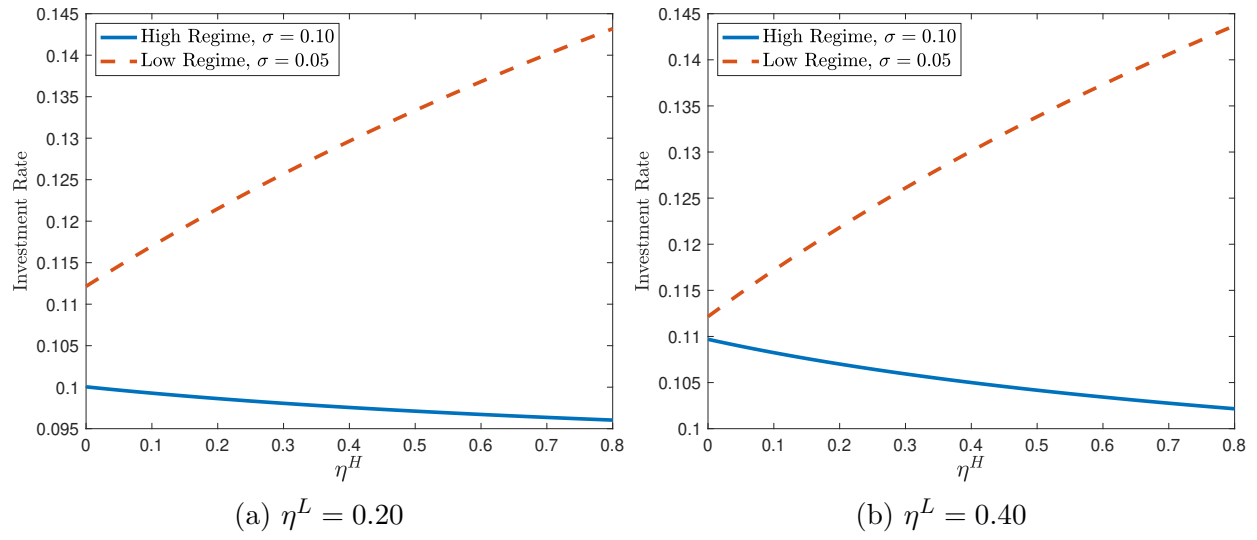


Figure IA.10: **Investment rate with stochastic volatility as a function of the arrival rate of the high regime η^H .**

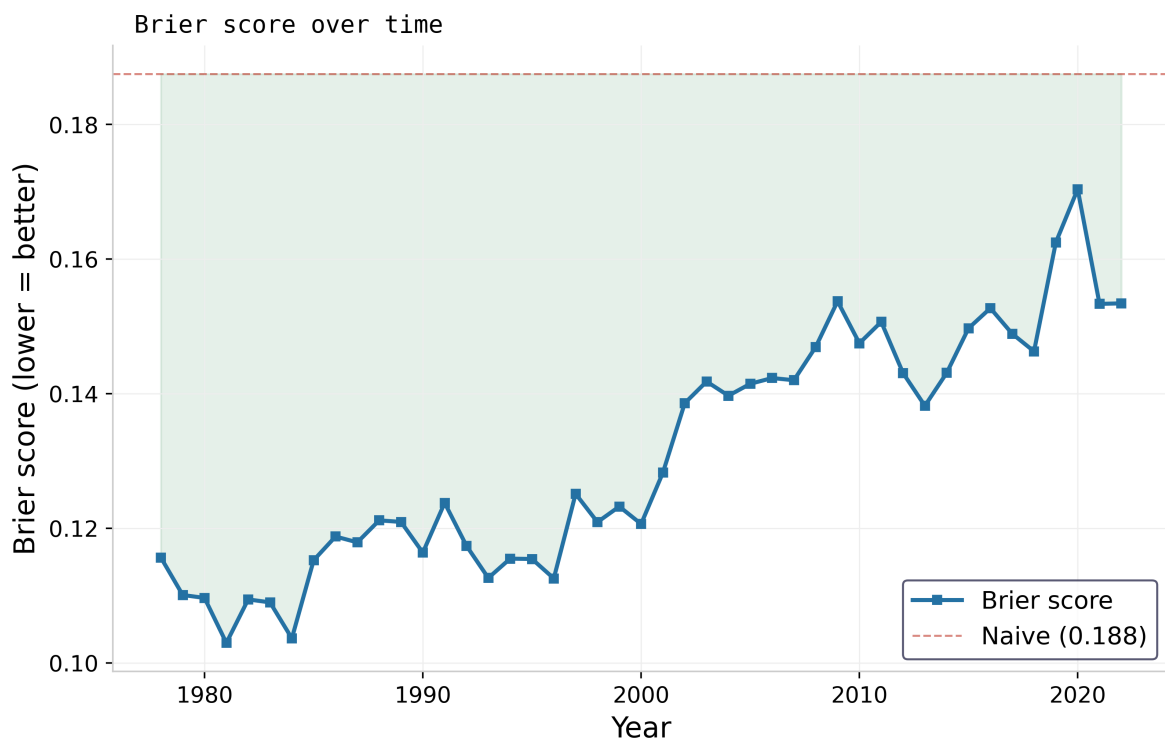
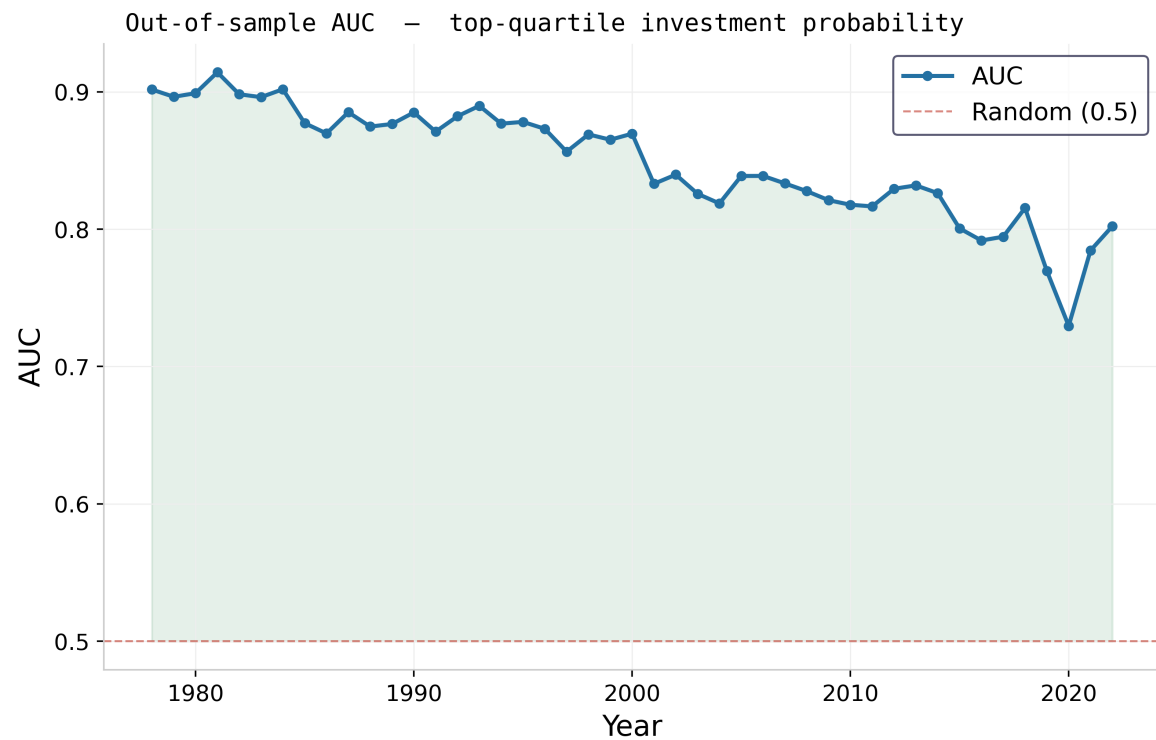


Figure IA.11: Diagnostics on Rolling Probit Regressions.